

Transposon proliferation drives genome architecture and regulatory evolution in wild and domesticated peppers

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Pepper (*Capsicum* spp.) is a widely consumed vegetable with exceptionally large genomes in Solanaceae, yet its genomic evolutionary history remains largely unknown. Here we present 11 high-quality *Capsicum* genome assemblies, including two gap-free genomes, covering four wild and all five domesticated pepper species. We reconstructed the ancestral karyotype and inferred the evolutionary trajectory of peppers. The expanded and variable genome sizes were attributed to differential transposable element accumulations, which shaped 3D chromatin architecture and introduced mutations associated with traits such as fruit orientation and colour. Using a chromatin accessibility atlas of *Capsicum*, we highlight the influence of transposable elements on regulatory element evolution. Furthermore, by constructing a haploblock map of 124 pepper core germplasms, we uncover frequent introgressions that facilitate the formation of sweet blocky pepper and the acquisition of important traits such as resistance to pepper mild mottle virus. These findings on the genomic and functional evolution of *Capsicum* will benefit pepper breeding.

Pepper belongs to the genus *Capsicum* of the family Solanaceae. With a cultivation area of 3.70 million hectares and a production value of US\$26 billion (2021, <http://www.fao.org/faostat/>), pepper is one of the world's most important vegetable crops. Peppers are primarily used as fresh vegetables and spices and are popular for their pungency and high nutritional value¹. Due to the physico-chemical properties of their secondary metabolites, such as capsaicinoids and carotenoids, peppers are also widely used in the production of various medicines, natural colours and cosmetics, as well as in raw materials for industrial coatings^{2,3}.

Capsicum comprises 5 domesticated pepper species and over 30 wild species^{4–7}. These *Capsicum* species are clustered into two main

groups: one is formed by the Andean clade, and the other includes the remaining species of the genus. The Andean clade may have had an independent origin and diverged earlier from the others in the *Capsicum* genus, a hypothesis that is supported by estimates of karyotype differences, genetic distance and geographical disjunction^{6,8,9}. Species in the Andean clade possess 13 pairs of chromosomes ($2n = 26$), in contrast to the most of the other *Capsicum* species, which have 12 pairs of chromosomes ($2n = 24$), a common configuration found in most diploid Solanaceae species^{9,10}. There is debate concerning the evolution of chromosome numbers in *Capsicum*, although most of the evidence suggests that species with 13 chromosome pairs were derived from species with 12 pairs^{11–14}. Additionally, species in the Andean clade

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have smaller genome sizes than other *Capsicum* species^{15,16}. The variation in karyotype and genome size indicates the complexity of genome evolution within *Capsicum*.

Evidence shows that the five domesticated pepper species—*Capsicum annum* L., *C. frutescens* L., *C. chinense* Jacq., *C. baccatum* L. and *C. pubescens* Ruiz & Pav.—were independently domesticated in different regions of Central and South America since at least 6000 BC^{4–7,17}. Among them, *C. annum* is the most widely cultivated one and has the highest economic value. The species includes many phenotypically diverse landraces that display variation in morphology, metabolites and yield¹⁸. In contrast, *C. frutescens* and *C. chinense* are mainly cultivated in certain countries or regions of America, Asia and Africa, while *C. baccatum* and *C. pubescens* are cultivated in Latin America^{4,16,19–21}. It is generally accepted that *C. annum*, *C. chinense* and *C. frutescens* belong to a small phylogenetic branch, showing parallel evolution for various plant and fruit characteristics^{22,23}. There are no interspecific barriers between the three species, and thus historical interbreeding may have occurred^{24,25}. However, how the genetic communication between these species has shaped their genomes and whether *C. chinense* and *C. frutescens* have promoted the diversification of *C. annum* are as yet unknown.

Due to the crucial role of genome sequences in guiding functional genomics and gene mapping studies, a few pepper genomes have been assembled using the short-read (eight genomes) or long-read (eight genomes) sequencing technologies^{26–33}. The majority of these assemblies were generated for the domesticated species, particularly *C. annum*. Genomic studies have shown that the genome size of domesticated peppers exceeds 3.0 Gb, approximately three to four times larger than that of closely related species such as tomato and potato. Comparisons between pepper and other Solanaceae species have revealed that the differences in transposable element (TE) content account for the majority of genome size variation^{34–36}. It is interesting to explore whether TE proliferation has played a role in the genome evolution and trait innovation of *Capsicum* species. Answering these questions requires a deep understanding of the genome composition of both wild and domesticated *Capsicum* species, a goal that cannot be achieved with the current limited number of high-quality genomes generated mainly for *C. annum*.

In this study, we assembled and analysed near-complete genomes of both wild and domesticated pepper species. We reconstructed the ancestral karyotype of pepper and inferred the trajectory of its karyotypic evolution. We show that extensive and differential TE accumulation not only shaped the genome architecture but also introduced mutations in genes associated with important traits in pepper. By constructing a chromatin accessibility map, we underscore the role of TEs in modulating gene expression regulation. Additionally, we found that recurrent introgressions accompanied the domestication of and trait evolution in *C. annum*. These findings offer valuable avenues for further study of pepper biology and breeding and provide insights into genome evolution in other plants and crops.

Results

Near-complete assemblies of wild and domesticated peppers

On the basis of the reported phylogeny¹⁸, we selected 11 pepper samples for genome assembly, including five wild and six domesticated peppers (Fig. 1a and Supplementary Tables 1 and 2). The five wild peppers included one *C. rhomboideum* from the Andean clade, one *C. chacoense* and one *C. eximium*, as well as two wild progenitors of *C. annum* (*C. annum* var. *glabriusculum*, PM1533 and Ac1979). The six domesticated peppers covered all five domesticated species, with two accessions showing the blocky ('Qiemen') and elongated (C5) fruit shapes for *C. annum* and one accession for each of the other four domesticated species, *C. frutescens*, *C. chinense*, *C. baccatum* and *C. pubescens*.

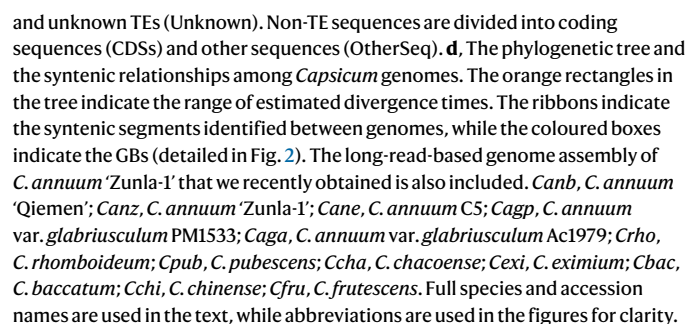
We generated both HiFi and ONT long reads for genome assembly. The average contig N50 and N90 lengths of these assemblies were 233.77 Mb and 116.86 Mb, respectively (Fig. 1b and Supplementary Table 3), indicating a high level of sequence continuity. Among these

assemblies, *C. rhomboideum* exhibited the smallest genome size (1.80 Gb), roughly half that of the other pepper species, as confirmed by flow cytometry (Supplementary Fig. 1). The genome sizes of the other assemblies ranged from 3.04 to 3.88 Gb (Fig. 1c). *K*-mer analysis revealed that the completeness of all assemblies exceeded 99% (Supplementary Table 3). We then generated Hi-C data (Supplementary Table 2) and assigned contigs to pseudo-chromosomes. Notably, two assemblies (*C. rhomboideum* and *C. annum* var. *glabriusculum* PM1533) reached the gap-free, telomere-to-telomere level, with all 24 telomeres fully assembled and no gaps remaining. Among the remaining assemblies, most (six out of nine) contained only six or fewer gaps, approaching a nearly gap-free level (Supplementary Table 3). These results indicate that the majority of pseudo-chromosomes (63.91%) were assembled as single contigs. The quality value scores of all assemblies were above 62 (Supplementary Table 3), meeting the standards for reference-grade genomes. Each assembly comprised 12 pseudo-chromosomes, except for *C. rhomboideum*, which had 13 (Fig. 1d). TE sequences occupied 79.41–85.55% of these genomes, and on average 37,489 protein-coding genes were predicted (Supplementary Table 3). The BUSCO values indicate high completeness in genome assembly (97.09–98.20%) and gene prediction (97.14–98.51%).

Karyotype-resolved evolutionary history of the *Capsicum* genus

We constructed a phylogenetic tree with tomato as the outgroup and revealed that *C. rhomboideum* from the Andean clade diverged first (approximately 9 million years ago (Ma)), and then the domesticated species *C. pubescens*, which represents the ancestral branch of left peppers (Fig. 1d). The remaining pepper species were divided into two clusters. Cluster one includes two wild species *C. chacoense* and *C. eximium*, and one domesticated species, *C. baccatum*; cluster two includes three domesticated species, *C. frutescens*, *C. chinense* and *C. annum*. The dispersed phylogeny of the five domesticated species indicates multiple parallel domestication events. Although *C. rhomboideum* showed a markedly smaller genome size and a large phylogenetic distance from the other peppers, a 1:1 syntenic gene relationship and well-conserved chromosomal synteny were observed between it and the others (Fig. 1d). Similar phylogenetic structures and synteny relationships were obtained by integrating the previously reported high-quality pepper genomes (Supplementary Table 1 and Supplementary Figs. 2 and 3), further validating our results.

The 13-chromosome Andean clade evolved from the 12-chromosome pepper ancestor. We selected the relatively ancestral genome *C. pubescens* as the representative of the 12-chromosome peppers and compared it with *C. rhomboideum* and other Solanaceae species, including golden berry, tomato and wolfberry, to identify chromosomal fission and fusion events on the basis of their genomic synteny (Supplementary Fig. 4). On this basis, we reconstructed the ancestral *Capsicum* karyotype (ACaK), which has 12 chromosomes (Methods). As shown in Fig. 2a, the ACaK contained 21 genomic blocks (GBs), including four GBs (C, J, K and O) that existed as four whole chromosomes across all investigated *Capsicum* genomes (Supplementary Table 4). Both *C. rhomboideum* and *C. pubescens* experienced three chromosomal reshuffling events (Fig. 2a and Extended Data Fig. 1). Interestingly, in *C. rhomboideum*, the ACaK chromosome A12 was split into two parts, which were subsequently translocated with A4 and A8, forming two new chromosomes. This result shows that the 13-chromosome genome of the Andean clade evolved from the 12-chromosome ancestor. In addition, the karyotype changes between different pepper species matched well with their phylogenetic relationships (Fig. 2a and Extended Data Fig. 2). For instance, species in cluster two experienced a common translocation event between chromosome 3 (Chr03) and Chr09, while the *C. annum* genomes feature an additional one between Chr01 and Chr08, leading to a longer Chr01 in *C. annum* than in *C. frutescens* and *C. chinense* (Extended Data Fig. 2).



nucleus, resulting in pronounced interaction signals in the Hi-C map³⁷. On this basis, we successfully identified the positions of centromeres. These centromeres aligned perfectly with the CENH3 ChIP-seq signals³³ (Supplementary Fig. 5). Comparison between *C. pubescens* and *C. rhomboideum* showed that a new centromere emerged in a

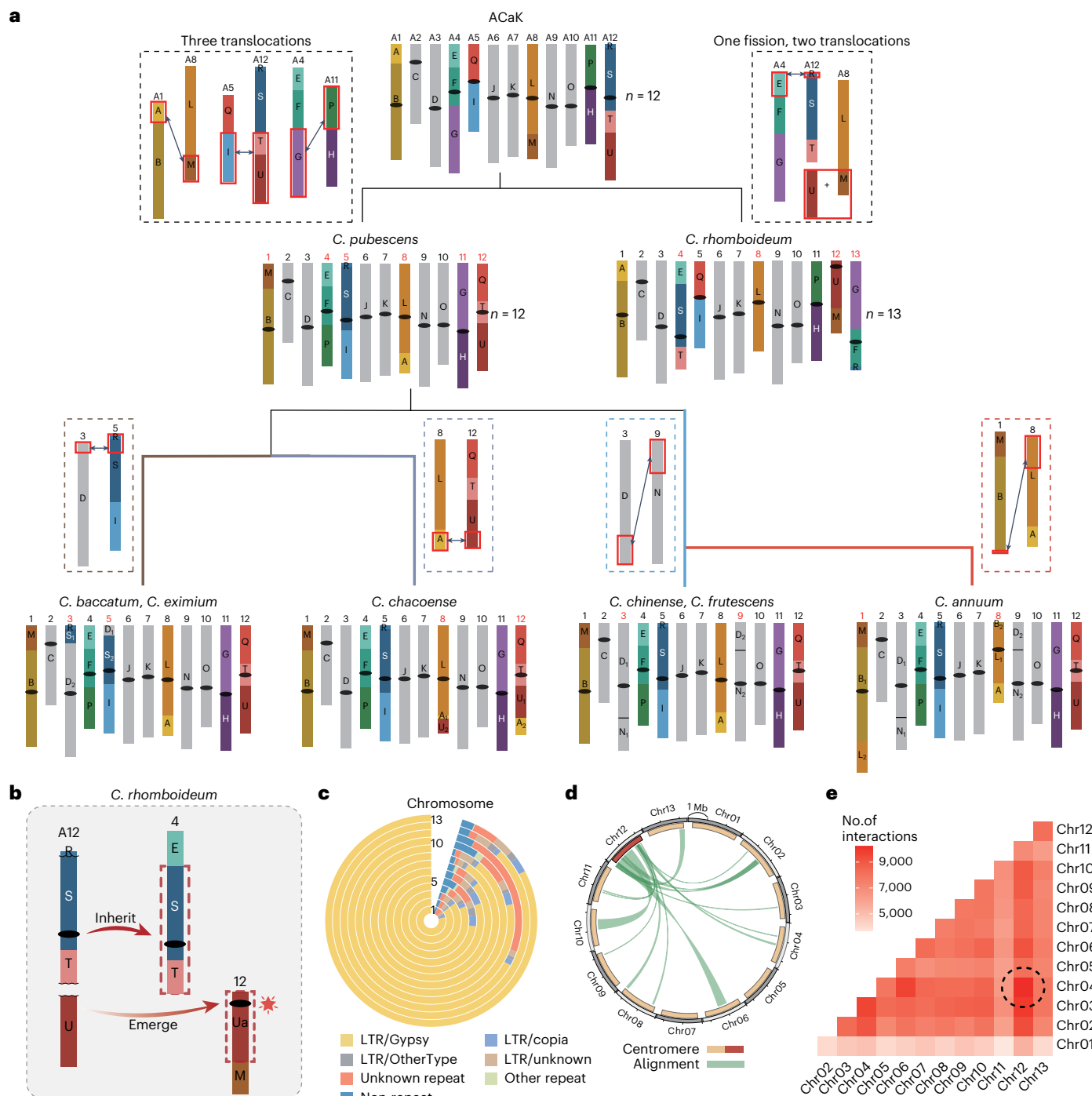


Fig. 2 | Karyotype and centromere evolution of *Capsicum*. **a**, The evolution of the karyotypes, which are represented by the composition of GBs indicated by different colours. The chromosomal rearrangement events are shown in the dashed boxes, with the involved genomic segments highlighted (red boxes). The chromosomes that experienced the rearrangement event are indicated in red. The number subscripts indicate the split GBs. The black ellipses denote

the centromeres. **b**, The emergence of a new centromere for Chr12 of the *Rho* genome. **c**, The sequence composition of the centromeres in *Rho*. **d**, Circos plot showing the sequence alignment (ribbons) between the centromere of Chr12 and all the other centromeres in *Rho*. **e**, Hi-C interaction map between the centromeres of *Rho*. The dashed circle highlights the strongest interaction in *C. rhomboideum* between Chr04 and Chr12.

repeat-rich region of GB U in the 13th chromosome of *C. rhomboideum* (Fig. 2b). GB U was ancestrally located in AcaK chromosome A12 (R-S-T-U), while chromosomal fission and translocation associated with GB M in the present Chr12 (Fig. 2b). The centromeres in Chr02 (GB C), Chr06 (GB J) and Chr10 (GB O) showed sequence similarity to the 13th centromere of *C. rhomboideum* (Fig. 2c,d). Interestingly, this 13th centromere showed the strongest Hi-C interaction with the centromere of Chr04 (Fig. 2e), which evolved from the same ancestral

chromosome; this is probably a relic of the intra-chromosomal interactions. Taken together, these results suggest that a chromosome fission event followed by the emergence of a new centromere accompanied the rise of the 13-chromosome karyotype of the Andean clade peppers.

Genomic variations associated with TE accumulation

The expanded genome size and its variation among pepper genomes were largely attributed to differences in TE accumulation, especially

the long terminal repeat-retrotransposons (LTR-RTs) (Fig. 1c). Among LTR-RTs, the *Gypsy* superfamily played a major role, with sizes varying from 664 to 1,293 Mb (Supplementary Table 5). To get a glimpse of the historical TE proliferation events, we estimated the insertion times of TEs by full-length LTR-RTs³⁸ (Fig. 3a,b, Extended Data Fig. 3 and Methods). The *Copia* full-length LTR-RTs in all *Capsicum* species except for *C. rhomboideum* showed an ancient proliferation (3.8 Ma) (Fig. 3b), which occurred after the divergence (9.0 Ma) between the Andean clade *C. rhomboideum* and other peppers. *C. rhomboideum* differs from other *Capsicum* species in the insertion pattern and strength of TEs. An insertion peak of the *Athila* family of *Gypsy* in *Cpub* coincided with the divergence time (1.8 Ma) between *C. pubescens* and other *Capsicum* species. We also observed a recent proliferation event of *Gypsy* (0.43 Ma), which was stronger in *C. baccatum* and *C. eximium* from cluster one, being correlated with their larger genome sizes. Species from cluster two showed an additional insertion peak of *Copia*, mainly those in the *Ale* family (Supplementary Table 5 and Extended Data Fig. 3). Meanwhile, these *Capsicum* genomes showed similar amounts of solo-LTRs, indicating that TE removal through homologous recombination functions similarly in these species (Supplementary Table 5). The LTR-RT scenario suggests that the differential accumulation of TEs from families such as *Athila* and *Ale* contributed to the diversity of genome expansion in peppers.

The recently inserted *Gypsy* LTR-RTs were significantly enriched in the centromeres of most *Capsicum* species (Fig. 3c). The majority of these *Gypsy* LTR-RTs were from the *CRM* family, which was known as centromeric retrotransposons in maize³⁹. However, this LTR-RT insertion pattern was absent in *C. annuum* (Fig. 3c,d, Extended Data Fig. 4 and Supplementary Fig. 6). Instead, *C. annuum* showed a higher density of *Gypsy* in gene-rich regions (Supplementary Fig. 7), further showing the different TE insertion landscapes among peppers. Interestingly, further comparison between *C. rhomboideum* and *C. pubescens* revealed a dramatic expansion of gene-syntenic regions (587 versus 1,733 Mb, 2.95-fold), which was stronger than the expansion of other regions—that is, the gene-sparse/centromere-proximal segments (1,130 versus 1,782 Mb, 1.58-fold; exemplified in Fig. 3c). In the syntenic regions, the expansion was associated with higher TE density and biased towards genomic regions with fewer genes (≤ 3 genes per 50 kb) (Fig. 3e and Supplementary Table 6), indicating the high plasticity of these regions.

Through aligning these pepper genomes to the genome of *C. annuum* var. *glabriusculum* PM1533, we identified 493,267 structural variations (SVs) using SyRI⁴⁰ (Methods), most of which were presence-absence variations (PAVs) (444,739) with sizes ranging from 50 bp to 493 kb. TEs occupied approximately 90% of these PAV sequences (Supplementary Table 7). This indicates that the frequent TE insertion/fractionation played a role in introducing sequence variation between pepper genomes, which might have affected the phenotypic diversity. For example, we previously identified the gene *Up* controlling fruit orientation, the variation of which was caused by a 579-bp TE sequence deletion in the *C. annuum* population¹⁸. Here we detected a new, larger deletion (8 kb) in the promoter region of *Up* in the

C. chinense assembled genome but not in the other *C. chinense* accessions (Fig. 3f,g). Intriguingly, the fruit orientation of this accession (erect) also differed from those of the other *C. chinense* accessions (pendent) (Fig. 3h). We validated the 8-kb deletion through PCR amplification (Extended Data Fig. 5) and found that *Up* expression was significantly higher ($P < 0.001$) in the pendent accessions than in the erect one in both the fruit pedicel and placenta, as revealed by an RNA-seq assay (Fig. 3i). These results indicate independent and parallel sequence deletions in the upstream region of *Up* in both *C. annuum*¹⁸ and *C. chinense* genomes, which were associated with the variation in expression of *Up*. Additionally, the gene *CCS* encodes capsanthin–capsorubin synthase, an important enzyme in carotenoid pigment biosynthesis. *CCS* was fine mapped and functionally validated to control the red fruit colour of peppers^{41–44}. We identified and validated 1.1-kb insertion of a *Copia* LTR-RT in the single coding exon of *CCS* in *C. pubescens*, correlating with yellow fruit colour rather than the typical red (Extended Data Fig. 6). This insertion caused a premature termination of *CCS*, as supported by RNA-seq data (Extended Data Fig. 6c).

Evolution of 3D chromosomal architecture

We identified the A/B compartments (Supplementary Table 8) and their switching events in *Capsicum* genomes using the Hi-C data (Methods). These genomes were spatially divided into two distinct compartments showing preferential interactions within types—the A and B compartments, which are associated with open and closed chromatin, respectively⁴⁵. Considering the relatively stable karyotypes and A/B compartments of species in the two clusters (Fig. 2a and Supplementary Table 8), we focused on A/B switching events between two representative genomes of *C. rhomboideum* and *C. pubescens*. Examination of these events in chromosomes that experienced rearrangements revealed that the changes in chromosomal positions, from distal to proximal or vice versa, were strongly associated with A/B compartment switching. For example, GB L was located in ACaK chromosome A08 with its left end as a telomere and its right end spanning the centromere (Fig. 2a). The relative position of GB L remained unchanged in *C. pubescens*, while the right end was exposed as a peritelomeric region due to a translocation in *C. rhomboideum*. Notably, the right end of GB L was predicted as a B compartment in *C. pubescens* but as an A compartment in *C. rhomboideum*, showing that the *C. pubescens*–B versus *C. rhomboideum*–A switching event was promoted by this translocation in *C. rhomboideum* (Fig. 3j). As another example, the ACaK chromosome A12 was composed of GBs R, S, T and U, with the R and the left end of S located in the peritelomeric region. However, in *C. rhomboideum*, the R–S GB association was translocated and linked with GB E (Extended Data Fig. 1). This event moved the R–S GB association away from the peritelomeric region to the pericentromeric region and resulted in the A-to-B switching in *C. rhomboideum* (Supplementary Fig. 8a). A similar phenomenon was observed in Chr12 of *C. rhomboideum* (Supplementary Fig. 8b).

To investigate the influence of genome expansion, we characterized the A/B switching events in chromosomes without ancestral

Fig. 3 | Evolution of pepper genomes, the fruit orientation trait and 3D architecture associated with TE proliferations and rearrangements.

a,b, The distribution of insertion times for *Gypsy* (**a**) and *Copia* (**b**) full-length retrotransposons. The line legends for the six reported high-quality *Capsicum* genomes, including three *C. annuum* (*ann*), are shown against a grey background. The blue box highlights the divergence between *Cpub* and other *Capsicum* species. **c**, Multiple genomic alignments between four genomes with Chr07 as an example. The ribbons indicate the syntenic segments. The dark blocks denote the centromeres. In the TE ratio (ii) and TE divergence (iii) rows, the orange, green and teal filled lines/dots denote the full-length *Copia*, *Gypsy* and other unknown types of TEs, respectively. The range of the TE divergence (iii) row is from 0 to 0.1, denoting the divergence between two LTRs. **d**, Distribution of the recently inserted (< 2 Ma) full-length *Gypsy* (fl-*Gypsy*). **e**, Genomic size expansion of *Cpub* compared to *Crho* (in each box plot, the centre line indicates the median, the box

limits indicate the first and third quartiles, and the whiskers extend to $1.5 \times$ the IQR). The *Crho* genes were categorized into 50-kb bins. **f**, The *Up* gene related multiple genomic alignments among *Capsicum* genomes. **g**, The location of the SV associated with *Up* in the *C. chinense* population. Deletion is denoted by the dashed box. **h**, Difference in fruit orientation of two *C. chinense* accessions with different *Up* genotypes (indicated by the black arrows). **i**, Expression level of *Up* in fruit pedicel and placenta samples of these two accessions determined using RNA-seq. The data are presented as mean $\pm$ standard deviation (biological replicates, $n = 3$; two-sided Student's *t*-test). Numbers above the bars indicate *P* values. **j,k**, Switching of A/B compartments (red boxes and ribbons) between the genomes of *Cpub* and *Crho*. Two types of chromosomes (Chr08 in **j** and Chr09 in **k**) are shown, representing those that experienced rearrangements and those that did not, respectively. Species and accession abbreviations are as defined in Fig. 1.

Most of the switched regions were clustered in the much-expanded interstitial regions (located between the peritelomeric and pericentromeric regions) (Fig. 3k and Supplementary Fig. 9), showing the high plasticity of these regions. Taken together, these results suggest that both chromosomal structure changes and TE-induced genome



expansion have promoted the evolution of *Capsicum* through altering the 3D genome architecture.

Regulatory elements rewired by TEs altered gene expression

To investigate the variation in regulatory elements of genes, we conducted assays for transposase-accessible chromatin sequencing (ATAC-seq) for the leaf samples from all 11 assembled *Capsicum* genomes and *C. annuum* ‘Zunla-1’, and we identified 46,171 to 62,142 accessible chromatin regions (ACRs) (Fig. 4a,b and Supplementary Table 10). As expected, ACRs were enriched upstream of genes, and 71% of ACRs were located within 5 kb of gene boundaries (Fig. 4a,c). The ACR sequences were highly conserved among *Capsicum* species. Taking *C. pubescens* as the reference, 81% of the ACRs in the analysed genomes, except for *C. rhomboideum*, were successfully aligned onto their syntenic genomic regions in *C. pubescens* (Extended Data Fig. 7a). The gene transcription strength was strongly associated with the distance between genes and adjacent ACRs (Fig. 4d). We further found a significant depletion of DNA cytosine methylations within ACRs, and this effect decreased with increasing distance between ACRs and adjacent genes (Fig. 4e and Extended Data Fig. 7b).

Although the genome size of *C. pubescens* was largely expanded relative to that of *C. rhomboideum*, the majority (74.41%, 2,762/3,712) of syntenic ACRs exhibited similar gene–ACR distances (less than twofold; Fig. 4f). For gene–ACR pairs with varied distances (at least twofold), most of them (83.55%, 320/383) were expanded in *C. pubescens*, and the size enlargement could be largely explained by the higher density of TEs (Fig. 4g). For example, a gene encoding ERF4 (a component bridging multiple hormone signal pathways) was surrounded by six conserved ACRs, and the distances between these ACRs and *ERF4* were significantly increased in *C. pubescens* due to TE insertions (Fig. 4h). We then expanded the comparison to all assembled pepper genomes and identified 76,521 accessibility-conserved ACR (acACR) groups across multiple (≥ 2) genomes (Fig. 4i and Methods). Among them, 8,131 acACR groups that were conserved in all genomes were proximal to genes enriched in fundamental biological functions, such as photosynthesis, phosphorus metabolic process and regulation of metabolic process (Extended Data Fig. 7c). Furthermore, through comparing the expression of syntenic orthologous gene pairs with or without acACRs in pairwise pepper genomes, we revealed that the gene pairs with acACRs tended to be expressed more concordantly than those without ($\text{PCC}_{\text{with}} - \text{PCC}_{\text{without}} = 0.1461$, where PCC represents the Pearson correlation coefficient, taking *C. annuum* ‘Zunla-1’ and *C. baccatum* as an example) (Supplementary Fig. 10). These results indicate the important regulatory role of acACRs in coordinating orthologous gene expression in different *Capsicum* species.

We identified 2,761 *annuum*-specific ACRs that were present in more than three *C. annuum* genomes but absent in all of the other *Capsicum* species. Genes with *annuum*-specific ACRs had great chances to be expressed more highly than their syntenic genes in other *Capsicum* genomes (Fig. 4j). For example, a gene encoding the phytochrome *PhyE* had an *annuum*-specific ACR. It was constantly expressed in *C. annuum* genomes but not in other species, except for *C. baccatum* (Supplementary Table 11). We found a TE-containing SV flanking this ACR between *C. annuum* and other *Capsicum* species (Supplementary Fig. 11). Moreover, an 8.8-kb TE was inserted into this ACR in *C. baccatum*, and a new ACR developed on this sequence debris. Correspondingly, *PhyE* was even more highly expressed in *C. baccatum* than in *C. annuum* ($P < 0.005$; Supplementary Table 11). This example suggests that the TE-associated SVs might be involved in the evolution of ACRs and gene expression regulation. In addition, *annuum*-specific ACRs exhibited significantly higher TE content than common ACRs ($P < 0.0001$, Fig. 4k). We also observed a negative correlation between ACR conservation and the TE ratio in ACRs ($\text{PCC} = -0.8402$, $P < 0.01$) (Fig. 4l and Methods). These results might reflect the influence of the high variability of TEs on the conservation of ACRs.

To investigate whether and how SVs affect the conservation of ACRs, we focused on two types of SVs: translocation SVs (tSVs), which lead to position changes of ACRs, and PAVs, which impair the structure of ACRs. For tSVs, we identified acACRs whose relative positions to adjacent genes were altered by tSVs, and we found that the accessibility of most of these ACRs was eliminated (Fig. 4m,n). For example, only 24% of *C. pubescens* tSV-associated ACRs remained accessible in *C. annuum* ‘Zunla-1’, while the ratio was 75% for ACRs without tSVs. For PAVs, we identified ACRs that showed size variation (>50 bp and $>20\%$ of ACR size) to assess the impact of PAVs. The accessibility level of these ACRs was lower than that of ACRs showing little size variation (68.67% versus 78.06%) (Fig. 4m,n). The decrease was more pronounced for those showing larger size variation ($>40\%$ of ACR size, 49.49%). Furthermore, we compared the PAV ratio in binned ACR sequences between ACRs that survived the variation and those that did not. Notably, the ACRs that survived exhibited much lower PAV ratios around the summit regions than around the distal regions and also than the ACRs that did not survive (Fig. 4o,p). This suggests the crucial role of ACR summits, the direct binding sites of transcription factors, over the flanking regions.

To investigate fruit ACRs, we selected the fruits of *C. pubescens*, *C. baccatum*, *C. chinense*, *C. annuum* var. *glabriusculum* PM1533 and *C. annuum* ‘Zunla-1’ and ‘Qiemen’ as representatives, on the basis of their phylogenetic relationships and differences in fruit traits. A total of 49,862–61,899 fruit ACRs were identified, and approximately 26–41% of these fruit ACRs were not detected in leaf samples (Supplementary Table 10). The fruit ACRs were less conserved between genomes than the leaf ACRs—that is, more fruit ACRs were found to be species-specific (Extended Data Fig. 7d), which might be associated with the distinctive fruit traits in different peppers. For example, *C. annuum* ‘Qiemen’ with its specific blocky fruit shape developed unique fruit ACRs in the promoter of *IQD21* and in the gene body of *IQD18*, both of which belong to a microtubule-associated protein family that play crucial roles in regulating fruit shape and seed germination through recruiting the Ca^{2+} sensor calmodulin^{46–49} (Supplementary Figs. 12a,b and 13). Both *IQD21* and *IQD18* had significantly ($P < 0.005$) higher expression in the fruit of ‘Qiemen’ than in other accessions with non-blocky fruit shapes (Supplementary Table 11). In addition, a ‘Qiemen’-specific fruit ACR was introduced by a TE insertion into the intronic region of *CIPK24*, an important gene involved in Ca^{2+} -dependent ion homeostasis⁵⁰ (Supplementary Fig. 12c). *CIPK24* exhibited significantly ($P < 0.0001$) higher expression in ‘Qiemen’ fruit than in others (Supplementary Table 11).

Introgressions increased the genetic diversity of *C. annuum*

To dissect the origin and genetic composition of the most cultivated and diverse *C. annuum* peppers, we generated deep resequencing data ($>40\times$ coverage) for a core collection of 124 pepper germplasms, including 81 *C. annuum* accessions (Supplementary Table 12)¹⁸. We used the IntroBlocker tool⁵¹ to reconstruct the haploblock map by evaluating binned divergence levels among accessions on the basis of identified variants (Methods).

We found that most of these introgressed haploblocks in *C. annuum* were from *C. frutescens* (207 Mb, 6.70% of the genome), while the *C. chinense*-derived haploblocks were rare (4 Mb, 0.13%) (Fig. 5a and Supplementary Table 13), showing a high frequency of genetic exchange between *C. frutescens* and *C. annuum*. The *C. frutescens*-introgressed haploblocks were mostly located at the peritelomeric regions (Supplementary Fig. 14a) and showed higher gene density (27 genes per Mb) than the *C. chinense*-related haploblocks (6 genes per Mb) (Fig. 5a and Supplementary Table 13). For most segments of these *C. frutescens* haploblocks (186/207), the genetic distances between *C. frutescens* and *C. annuum* were significantly greater than those between *C. annuum* accessions ($P < 0.05$) (Supplementary Fig. 14b), suggesting that these haploblocks were introgressed from *C. frutescens* into *C. annuum* rather than the other way around. Furthermore, among the divergent haploblocks (803) between two assembled wild progenitors of *C. annuum*

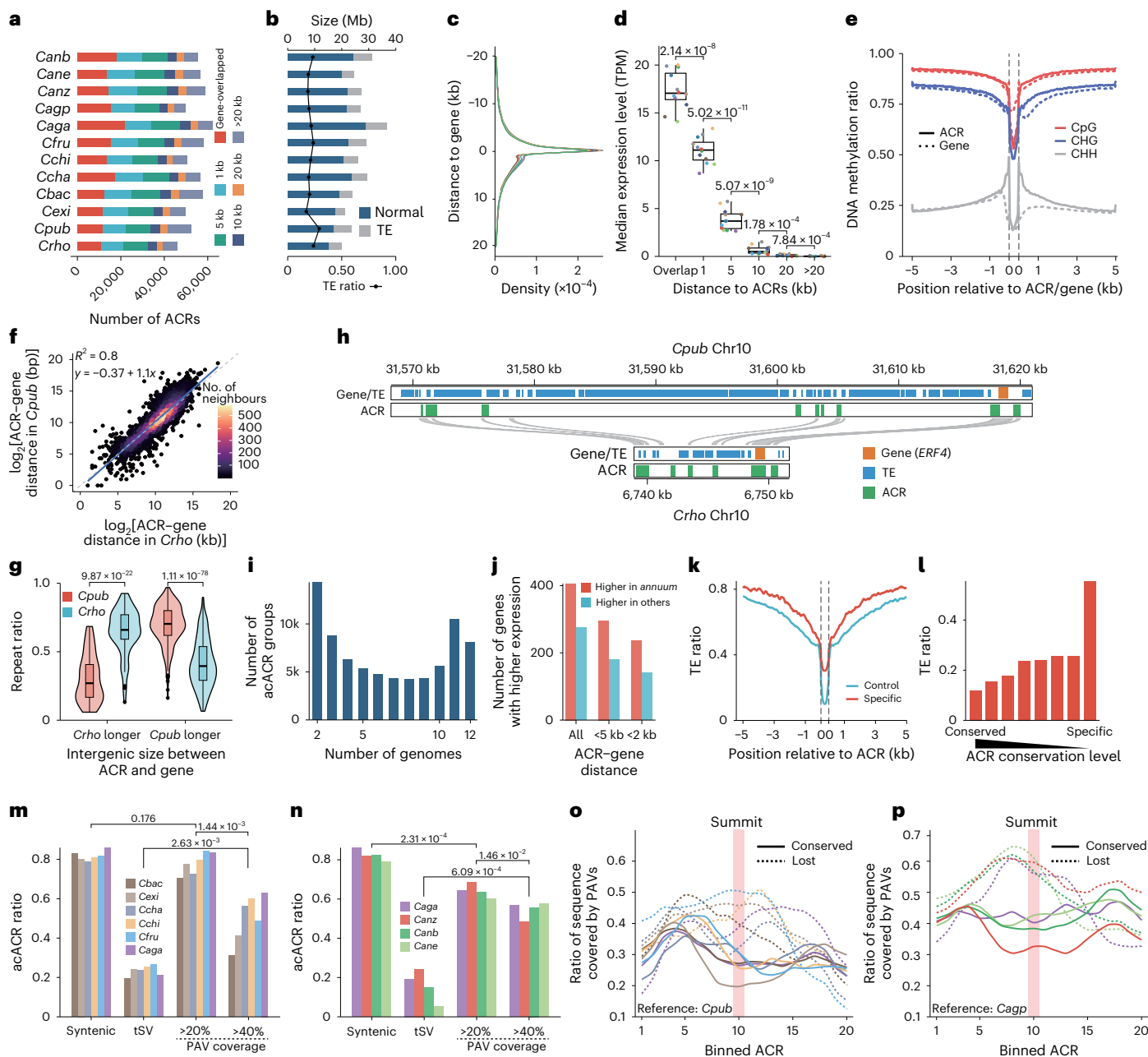


Fig. 4 | Chromatin accessibility map of *Capsicum* and associations with TEs and SVs. **a**, Number of ACRs at different distances from adjacent genes. **b**, Total size and TE ratio of ACRs. **c**, Distance distribution of ACRs to the translation start codon of the nearest genes. **d**, Median expression levels of genes with ACRs at different distances ($n = 12$; two-sided Student's t -test; in each box plot, the centre line indicates the median, the box limits indicate the first and third quartiles, and the whiskers extend to $1.5 \times$ the IQR). TPM, transcripts per million. **e**, DNA methylation levels in and around ACRs and genes, using the *C. annuum* 'Zunla-1' genome as an example. **f**, Comparison of ACR-gene distances between *Cpub* and *Crho*. **g**, Repeat ratios of intergenic sequences between genes and ACRs with varied ACR-gene distance in *Cpub* and *Crho*. '*Crho* longer' indicates $\geq 2 \times$ longer distances in *Crho* than in *Cpub* ($n = 63$), while '*Cpub* longer' indicates the opposite ($n = 320$) (two-sided Student's t -test; in each box plot, the centre line indicates the median, the box limits indicate the first and third quartiles, and the whiskers extend to $1.5 \times$ the IQR). **h**, Example of ACR conservation in regions expanded in

Cpub compared with *Crho* for an ERF4 gene. **i**, Distribution of acACR groups in different numbers of *Capsicum* genomes. **j**, Genes with *annuum*-specific ACRs generally exhibit higher expression than their syntenic genes without ACRs in other *Capsicum* species. **k**, TE contents in and around *annuum*-specific ACRs and those not specific to *C. annuum* (control). **l**, TE contents in ACRs conserved in different numbers of genomes. **m, n**, Ratios of conserved ACRs when *Cpub* (**m**) or *Cagg* (**n**) was set as a reference (two-sided Student's t -test). ACRs were categorized into three types: those located in syntenic regions (Syntenic), those with location changes but still conserved in sequences (tSV) and those located in syntenic regions but showing PAVs (>50 bp). **o, p**, PAV-covered sequence ratios of two types of ACRs, shown in bins. The solid lines indicate ACRs retaining accessibility; the dashed lines indicate ACRs with lost accessibility. The line colours follow the scheme in **m** and **n**. Numbers above the bars and box plots represent P values. Species and accession abbreviations are as defined in Fig. 1.

(PM1533 and Ac1979), haploblocks more similar to those in PM1533 were found in a larger number of *C. annuum* cultivars (19 cultivars on average) than those more similar to Ac1979 (9 cultivars on average) (Fig. 5b). This suggests that the wild progenitor group represented by,

or closely related to, PM1533 probably had a broader genomic influence on *C. annuum* cultivars than the group represented by, or closely related to, Ac1979. In addition, the specific haploblocks of the blocky-fruit pepper were limited, in contrast to the considerable haploblocks specific to

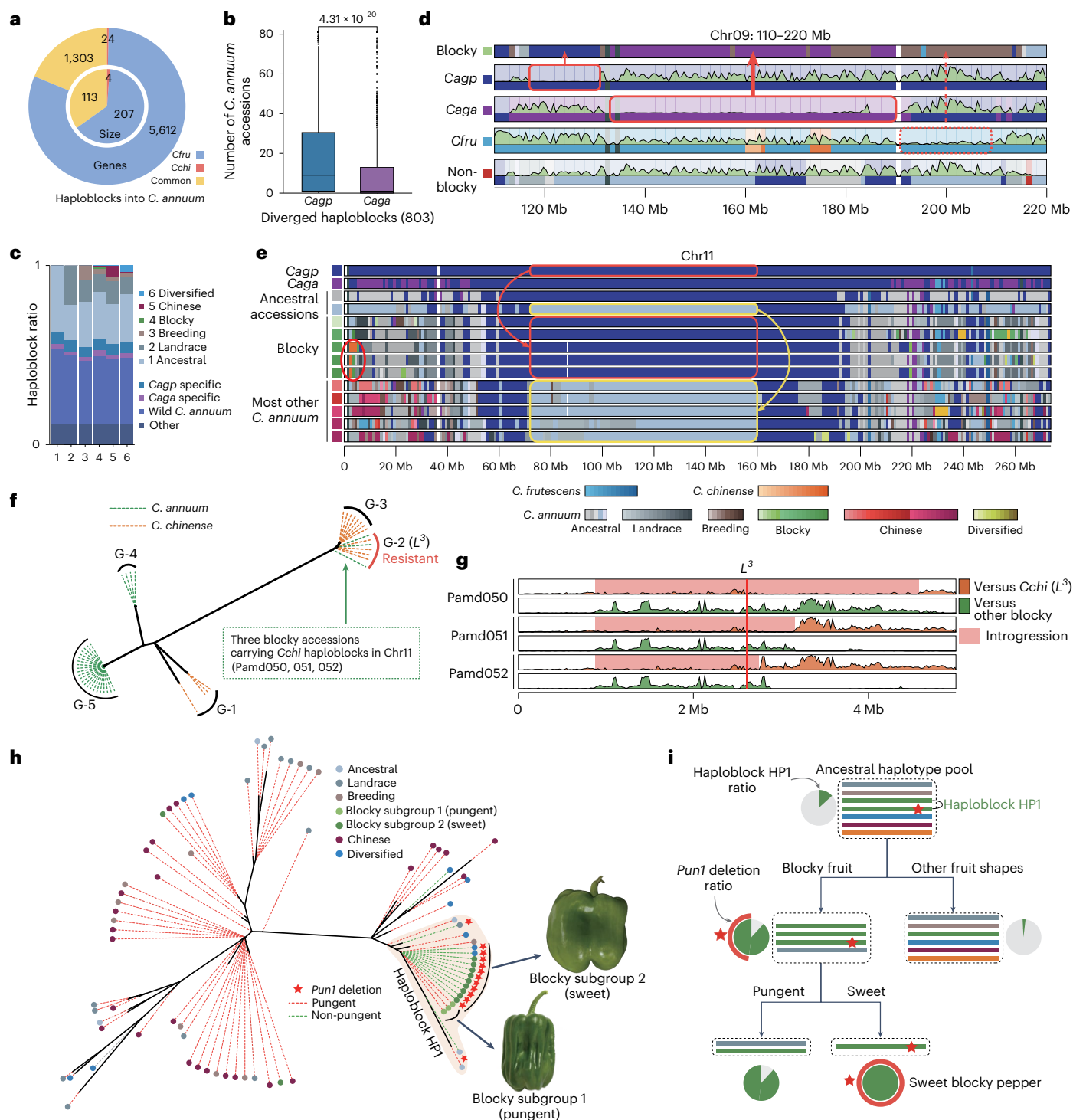


Fig. 5 | Haploblock scenario of *C. annuum* and the evolution of resistance- and pungency-related haploblocks. **a**, Contributions of *Cfr* and *Cchi* haploblocks to *C. annuum*. Ancestral undifferentiated haploblocks (common) are shown in yellow. **b**, Contributions of diverged haploblocks from *Cagp* or *Caga* to the domesticated *C. annuum* population ($n = 380$ versus 194; two-sided Student's *t*-test; in each box plot, the centre line indicates the median, the box limits indicate the first and third quartiles, the whiskers extend to $1.5 \times$ the IQR and the data points indicate outliers). Numbers above the bars indicate *P* value. **c**, Haploblock composition in six *C. annuum* groups based on the pepper variome (Supplementary Table 12). **d**, Centromeric region of Chr09 in blocky-fruited *C. annuum* (represented by Pamd032) contains haploblocks similar to those in *Caga*, *Cagp* and *Cfru* (represented by Pfru003). The coloured boxes denote the haploblocks, while the solid lines denote the genetic distances between Pamd032 and the other accessions. Recent and ancient introgressions are indicated by solid and dashed arrows, respectively. **e**, Haploblock diversity in Chr11. The evolutionary trajectories

of haploblocks are highlighted by red and yellow boxes and arrows. The *Cchi*-derived haploblocks (orange) in three blocky accessions are marked by red circles. **f**, Phylogenetic tree of the *L* gene in *C. annuum* and *Cchi* accessions. Among the five clades, which correspond to five genotypes (G-1 to G-5), G-2 is the *L*³ genotype, with resistance to PMMoV. **g**, Introgressed segments (light red) in three blocky accessions that were deduced according to the genetic distances against the *L*³-carrying *Cchi* accession (orange) and other blocky accessions (green, control). **h**, Phylogenetic tree of the haploblock (152–153 Mb of Chr02) carrying *Pun1*. Group information for each accession is indicated by coloured circles. Pungent and non-pungent accessions are indicated by red and green dashed lines, respectively. Haplotypes carrying the deletion of *Pun1* (red star) belong to a specific haploblock (light yellow background, named HP1). **i**, Evolution of *Pun1*-carrying haploblock in *C. annuum*. Different haploblocks are represented by coloured bars for illustrative purposes. The pie charts indicate the ratios of haploblock HP1 (green) in the populations. Species and accession abbreviations are as defined in Fig. 1.

the previously defined Chinese pepper groups¹⁸ (Fig. 5c). Intriguingly, the blocky pepper had a higher proportion of haploblocks with high similarity to Ac1979 than the other *C. annuum* cultivars did, indicating that the recent emergence of blocky peppers involved additional contributions from some wild progenitor of *C. annuum*.

The haploblocks located in the peritelomeric regions showed high transition frequencies (Extended Data Fig. 8), while those in the pericentromeric and centromeric regions were relatively long and consecutive. With the assistance of these consecutive centro-haploblocks, we were able to identify the large genomic regions that diverged between different samples and to trace their origins, such as the regions in Chr02, Chr09 and Chr11. Among them, the centro-haploblocks in Chr09 and Chr11 corresponded to the reported fragments F9 and F11 in blocky pepper¹⁸, respectively. We here reveal that the series of centro-haploblocks in Chr09 (117–215 Mb) is composed of multiple genetic sources rather than a simple introgression from wild *C. annuum*. The left part shows higher similarity to the haploblocks of PM1533 than to those of Ac1979, the middle part shows higher similarity to those of Ac1979 and the right end probably represents the remnants of historical introgression from *C. frutescens*, with accumulated mutations over time (Fig. 5d). The combination of these haploblocks was highly conserved in the blocky peppers but was almost non-existent in other *C. annuum* samples (Supplementary Fig. 15). Meanwhile, a series of centro-haploblocks in Chr11 (72–160 Mb) was also conserved (named H1) in blocky pepper but was limited in others (Fig. 5e). Unexpectedly, H1 was inherited from the wild *C. annuum* (both PM1533 and Ac1979), suggesting that another combination of haploblocks (H2) that was dominant in non-blocky *C. annuum* was the derived haplotype. H2 became the major one, whereas H1—reduced to a minor haplotype—accompanied the divergence of blocky pepper. In addition, the ancestral centro-haploblock in Chr02 (44–54 Mb) was almost fixed in blocky pepper but rarely occurred in others, especially not in the Chinese pepper groups¹⁸ (Extended Data Fig. 9). These results show that the divergence and formation of *C. annuum* cultivars were coupled with frequent genomic introgressions from wild *C. annuum*.

Important genes previously reported to be associated with disease defence and non-pungency traits occur in these haploblocks dominant in blocky peppers (Supplementary Table 14). We observed that three blocky pepper accessions carried the *C. chinense*-derived haploblocks in Chr11, whereas none of the other *C. annuum* accessions possessed these haploblocks (Fig. 5e). Intriguingly, within these haploblocks, we identified a CC-NBS-LRR gene named *L*, known to confer resistance to *Tobamovirus* spp.⁵². The *L* gene introgressed from *C. chinense* was the *L*³ allele, one of the four known alleles (*L*¹–*L*⁴). The *L*³ allele has been validated to mediate resistance against the *P*₀, *P*₁ and *P*_{1,2} pathotypes of pepper mild mottle virus (PMMoV)^{53–55}. Both haploblock tracing and phylogenetic analysis suggested that the *L*³ allele originated from *C. chinense* accessions (Fig. 5e,f), while the sizes of these introgressed segments in these three accessions were determined to be from 1.87 to 3.70 Mb (Fig. 5g). The blocky pepper and the *C. chinense* accessions harbouring the *L*³ allele showed strong resistance to the *P*_{1,2} pathotype as revealed by the inoculation test (Extended Data Fig. 10a). It is noteworthy that *L*³ was located in the present genotype of a 493-kb PAV (Extended Data Fig. 10b), which might be associated with its rare appearance in the *C. annuum* population. Non-pungency is one of the most important traits of sweet blocky peppers. In the genomic region of the well-known *Pun1* gene, we observed a blocky-pepper-dominant haploblock. *Pun1* encodes a key enzyme in capsaicin biosynthesis, and a loss-of-function mutation in this gene results in non-pungent peppers⁵⁶. This blocky-pepper-dominant haploblock existed at a low frequency (19%) in the ancestral and landrace accessions of *C. annuum*. The deletion of *Pun1* occurred and resulted in the polymorphism of this specific haploblock. Although this haploblock was nearly fixed in all blocky peppers, the *Pun1* deletion genotype was consistently present in sweet blocky peppers but absent in pungent blocky peppers (Fig. 5h).

This suggests that the inheritance of this haploblock from ancestral *C. annuum* accompanied the formation of blocky pepper, while the selection of the non-pungency trait for sweet blocky peppers led to the widespread presence of the *Pun1* deletion allele in this haploblock (Fig. 5i). These results together indicate that these haploblocks specifically retained in blocky pepper contain gene candidates that contribute to the evolution and domestication of blocky pepper.

Discussion

We generated chromosome-level genome assemblies for both wild and domesticated peppers with varied genome size. On the basis of these 11 newly assembled and eight individually reported high-quality pepper genomes, we reconstructed the ancestral karyotype of *Capsicum*, inferred the evolutionary trajectory of pepper genomes, and characterized the large fragment rearrangements and frequent introgression events, as well as the evolution of the 3D chromosomal architecture. We also revealed the genetic diversity of genes and regulatory elements shaped by SV during the speciation and domestication of *Capsicum*.

Compared with most diploid Solanaceae species, the much-expanded genome size of *Capsicum* has raised questions concerning how this pronounced genome expansion occurred and what the evolutionary and genetic consequences are. Previous studies comparing *C. annuum* with other Solanaceae species suffered from the effects of large phylogenetic distances^{26,27} and the limited number of high-quality pepper genomes. The wild *C. rhomboideum* assembled here shows a significantly smaller genome size (1.8 Gb), and it diverged from the ancestor of *C. pubescens* approximately 9.2 Ma, much shorter than the distance between *Capsicum* and tomato (16.0 Ma) or golden berry (15.5 Ma)⁵⁷. Comparisons between *C. rhomboideum* and other pepper genomes revealed differences in TE accumulations that underlie genome size expansion and variation among *Capsicum* species, which could potentially be attributed to TE proliferation events, as well as other factors such as selective pressures and TE elimination. The activation of different TE families might be related to historical changes in ecological or geographical environmental factors, accidental inter-specific hybridization (genomic shock) or other factors⁵⁸. Although the rapid transposition of selfish TEs frequently introduced deleterious variations, the *Capsicum* species that survived may have benefited from TE transposition. The TE-involved elongation of the interstitial regions and the chromosomal rearrangements affected the 3D genome architecture of pepper. Meanwhile, TE proliferations also introduced variations proximal to genes. These variations enriched the genetic diversity of genes controlling important traits, thereby contributing to phenotypic diversity in pepper.

Much effort has been devoted to investigating the biological consequences of SVs, most of which are associated with TE transposition^{59–62}. It is now acknowledged that SVs are able to exert their functions through inducing functional mutations, generating variation in copy number and manipulating gene expression^{61,63}. However, little is known about the influence of TE-involved SVs on chromatin accessibility. We compiled a chromatin accessibility map for 12 *Capsicum* accessions, whose genomes showed size variation due to differential TE insertion. We demonstrated that the evolution of ACRs was closely related to TEs. Interspecies comparisons showed that SVs, both the macro type affecting the location and the micro type altering the sequence composition of ACRs, strongly affected the function of ACRs. These results suggest a new approach to how SVs are involved in the regulation of gene expression in plants and thus expand our knowledge of SV-mediated functional innovation.

Identifying the major factors involved in shaping the present *Capsicum* genomes and traits not only facilitates the interpretation of pepper species evolution but also inspires potential breeding strategies for pepper. Through building the haploblock landscape for *C. annuum* and sister species, we identified frequent introgression events between peppers. Introgressions from *C. frutescens* to *C. annuum*

might have facilitated the spread of TEs located in the introgressed segments (Supplementary Fig. 16), which could contribute to the genetic variations in the *C. annuum* population. We observed a high frequency of haploblock transitions and diverse haploblock combinations in the peritelomeric regions of *C. annuum*, suggesting that *C. annuum* may possess relatively high genetic diversity. Meanwhile, we uncovered the unbalanced contributions of sister species *C. frutescens* and *C. chinense* to the genetic repertoire of the current *C. annuum* cultivar population. These results suggest that the introduction of underutilized genetic materials such as *C. chinense* into *C. annuum* is a promising approach for increasing the haplotypic and thus phenotypic diversity of *C. annuum*.

This is particularly important for tracing the origins of haploblocks in pepper germplasm, especially the elite breeding materials. Due to the reduced recombination rates in the pericentromeric regions, the centro-haploblocks were capable of being unambiguously distinguished. These fossil-like haploblocks provided an excellent opportunity to trace the domestication and breeding history of pepper. We thereby revealed the major ancestral donor and determined how the haploblocks were transmitted along with the diversification of pepper. For example, the reference *C. annuum* genome ‘Zunla-1’ exhibited a rare haploblock combination in the pericentromeric region of Chr02, shared with only two accessions (Pamd012 and 013), indicating potential historical hybridization among these accessions. Another example is the introgressed *L*³-carrying haploblock from *C. chinense* into *C. annuum*, which probably introduced PMMoV resistance, although we cannot completely rule out the possibility that other loci within this introgressed fragment may also contribute to this resistance. We anticipate that with the accumulation of more resequencing data—particularly from *C. annuum*, *C. annuum* var. *glabriusculum*, *C. chinense* and *C. frutescens*—the deduced haploblock landscape will provide a more comprehensive understanding of the haploblock origins of *C. annuum* cultivars and play an increasingly important role in directing pepper breeding via the use of previously neglected genomic resources.

We dissected the evolutionary trajectory of *Capsicum* species from two major aspects: the evolution of the genome architecture and the haploblock introgression/inheritance. We constructed a chromatin accessibility map for wild and domesticated peppers and described the regulatory variation of genes accompanying pepper genome evolution. Both the large-scale genomic and the micro-scale regulatory variations identified in this study highlight the importance of extensive and differential TE proliferation. These results provide a comprehensive evolutionary roadmap of *Capsicum* species that will benefit future breeding efforts, and they greatly deepen our understanding of how genome expansion leads to genomic and gene functional/regulatory evolution in plants.

Methods

Plant materials and genome sequencing

The accessions for genome assembly were selected from our core collection at the Institute of Vegetables and Flowers in Beijing, China. The materials were planted in the greenhouse with at least three replicates. Young leaves of each accession were collected and then flash-frozen in liquid nitrogen. DNA was extracted from the leaves using the CTAB method. The PacBio HiFi libraries were constructed using the SMRTbell Express Template Prep Kit 2.0 and were sequenced on PacBio HiFi Revio or Sequel II sequencing platforms (Pacific Biosciences). The ONT libraries were constructed using the Ligation Sequencing Kit 1D (PM) and were sequenced on an Oxford Nanopore PromethION sequencer (Oxford Nanopore Technologies). Young fresh leaves of each accession were sampled for the construction of Hi-C libraries, which was then conducted following standard protocols using the Proximo Hi-C Plant Kit. The HindIII enzyme was used for treating the DNA. The libraries were sequenced on the Illumina HiSeq2000 platform. The 150-bp

paired-end short reads were generated on the Illumina NovaSeq or DNBSEQ-T7 platform.

Genome assembly and scaffolding

The genome size and heterozygosity of each accession were estimated using Jellyfish (v.2.2.10) and GenomeScope (v.2.0) with the *k*-mer size set as 31 (ref. 64). Hifiasm (v.0.20.0) was used to assemble the HiFi reads, ONT reads and Hi-C reads into the haplotype-phased contigs with the parameters –dual-scaf; –telo-m, AAACCT⁶⁵. For the *C. annuum* C5 without ONT data, only HiFi and Hi-C reads were used for assembly. The Hi-C reads were aligned to the assembled contigs using Chromap (v.0.2.6)⁶⁶, and the resultant alignment SAM files were sorted using SAMtools (v.1.16). We then used YaHS (v.1.1) for scaffolding the contigs with the parameter no-contig-ec⁶⁷. Juicer (v.1.6)⁶⁸ and Juicebox (v.2.18)⁶⁹ were used to generate and visualize the interaction heat map, respectively, and the potential misassemblies were manually corrected. Alternatively, ONT-only contigs were generated using NextDenovo (v.2.4.0)⁷⁰. The Nanopore-based contigs were further polished using NextPolish (v.1.3.0) with the assistance of short reads. The ONT-based contigs were aligned to the scaffolds of the assembled genomes using Nucmer (v.4.0.0) to identify potential misassemblies in these scaffolds⁷¹. SVs were manually identified, and the corresponding local Hi-C heat maps were revisited. To close the gaps remaining in the scaffolds, contigs of haplotypes and the ONT-based contigs were aligned to the scaffolds using Nucmer, and gaps were manually closed by replacing the gaps with the corresponding sequences from these contigs. The telomeric repeat motif AAACCT was screened using Tandem Repeats Finder (v.4.09)⁷². The completeness of the assemblies was assessed using Merqury (v.1.3) from HiFi reads with the *k*-mer size set as 31 (ref. 73). BUSCO (v.3.0.2) analysis was performed with the embryophyta_odb 10 database.

Genome annotation

RepeatModeler2 (v.2.0.4) with ‘LTR’ was used to generate a non-redundant repeat library for each accession⁷⁴. For the library, RepeatMasker (v.4.1.5) was used to identify and mask the repetitive sequences⁷⁵. LTR_retriever (v.2.8), which integrates the results from LTR_Finder and LTR_harvest, was used to better annotate the full-length LTR-RTs and to measure the pairwise divergence rate (*K*) between two LTRs of each retrotransposon^{76–78}. The insertion times of these full-length LTR-RTs were estimated according to the formula $K/2r$, with a substitution rate (*r*) of 6.0×10^{-9} , which was set on the basis of the divergence times between *Capsicum* species. The full-length LTR-RTs were further classified into families using TEsorter (v.1.4.6)⁷⁹. Protein-coding genes were predicted on the basis of the gene models from ab initio, homologue-assistant and transcript-based approaches. The detailed methods were described in our previous study⁸⁰. In brief, we generated the mRNA-seq data (approximately 6 Gb of Illumina paired-end data per sample) for leaves, roots, stems, fruits and flower organs from each accession. HISAT2 (v.2.2.0) followed by StringTie (v.2.1.0) and Trinity (v.2.12.0) were used to obtain the genome-assistant and ab initio transcript models, respectively^{81–83}. Protein sequences of 12 species, including *Arabidopsis thaliana*, coffee, cacao and several Solanaceae species (tomato, potato and petunia) were collected for homologue-based prediction using Exonerate (v.2.2.0). The ab initio gene predictions were generated using Augustus (v.3.5.0) and GeneMark-ES (v.4.72), both of which included the results from the RNA-seq data^{84,85}. EvidenceModeler (v.1.1.1) was used to integrate the gene models, and genes with incomplete structure were removed⁸⁶. In addition to the newly assembled genomes, the previously released genome assemblies were also annotated to ensure comparability. The predicted proteins were aligned to the SWISS-PROT database (<https://www.uniprot.org>) with $E < 1 \times 10^{-5}$, and the annotations of the top five hits were retained⁸⁷. The proteins were also submitted to the InterPro database (<https://www.ebi.ac.uk/interpro/>) to annotate the functional domains and to assign the Gene Ontology terms⁸⁸.

Identification of syntenic gene families

Syntenic gene pairs were identified across tomato, potato, physalis and 13 pepper genomes using mSynOrths (v.0.1; <https://gitee.com/zhanglingkui/msynorths>) with the parameters $m = 0.7$, $x = 10$ and $n = 5$. Specifically, a homologous gene pair was defined as syntenic if its flanking genes were also homologous. Two syntenic pairs interrupted by fewer than 50 genes were grouped into a single syntenic segment. A valid syntenic fragment requires at least 10 gene pairs. Syntenic gene pairs identified between species were then clustered into syntenic gene families across all investigated species.

Phylogenetic analyses

Syntenic families containing members from all of the investigated species and accessions were used to construct a phylogenetic tree. Protein sequences of syntenic gene families were aligned with mafft (v.7.495)⁸⁹ using the default parameters. The alignments were trimmed with trimAl⁹⁰ (v.1.4.1; parameter automated1), and the trimmed sequences of each species or accession were concatenated into one sequence. On the basis of the concatenated sequence, a phylogenetic tree was constructed using RAxML⁹¹ (v.1.1.0; parameters -f, a; -N, 100; -m, PROTGAMMAJTT). The divergence times were inferred using the Compute Timetree function of MEGA X⁹², and several known divergence times obtained from Timetree (<http://www.timetree.org/>)⁵⁷ were used as references to calibrate the divergence times among *Capsicum* accessions.

K_s analysis

Protein sequences of homologous gene pairs were aligned using ParaAT (v.2.0)⁹³. Protein alignments were then converted into coding sequence alignments using an in-house Perl script (available upon request). Synonymous (K_s) substitution rates were calculated on the basis of the coding sequence alignments using the method of Neil and Gojobori implemented in the KaKs_calculator (v.3.0)⁹⁴.

Reconstruction of the ancestral karyotype

Protein sequences from *C. pubescens* and *C. rhomboideum* were aligned to those of diploid Solanaceae species, including physalis, tomato and wolfberry. Rearrangement-related break points were identified through interspecies comparisons. Shared break points and corresponding genomic segments between *C. pubescens* and *C. rhomboideum* were considered ancestral before their divergence, while species-specific break points were considered independently generated post-divergence. To reconstruct the ACAk, the combination of genomic segments was optimized by minimizing necessary rearrangements. Segments unchanged during the speciation of *C. pubescens* and *C. rhomboideum* were labelled as GBs A to U according to their order in the ACAk. Using the ACAk system, rearrangements introduced during the evolution from *C. pubescens* to other *Capsicum* species were deduced through synteny comparisons.

Centromere identification and comparison

Centromeres were identified on the basis of their exceptionally strong interaction in the Hi-C map. We used FAN-C (v.0.9.0) to calculate the interaction frequency between genomic bins with sizes ranging from 5 kb to 5 Mb (ref. 95). Potential centromere-related bins were roughly framed by selecting bins with the strongest inter-chromosomal interactions. Interaction heat maps of these bins across chromosomes were visualized, and the boundaries of these interaction hotspots were manually refined according to their interaction pattern. The exact centromeric boundaries in each chromosome were determined by averaging the positions from all-versus-all inter-chromosomal comparisons. We used publicly available CENH3 ChIP-seq data for *C. annuum* (PRJNA962192, National Center for Biotechnology Information) to validate the positioning of centromeres. Considering that this ChIP-seq dataset was generated for a domesticated *C. annuum* accession, the ChIP-seq reads were aligned to the *C. annuum* 'Zunla-1' genome

using Bowtie2 (v.2.4.4). The bamCoverage function in deepTools (v.2.0) was used to calculate the normalized depth relative to the input sample with the parameters -of, bedgraph; -operation, ratio; -binSize, 100000; -smoothLength, 300000; -normalizeUsing, CPM; -scaleFactorsMethod, None⁹⁶. Centromeric regions predicted in *Capsicum* genomes were extended 5 Mb upstream and downstream to evaluate their repeat composition.

Identification of SVs

Each of the assembled *Capsicum* genomes was aligned to the PM1533 genome using minimap2 (v.2.24-r1122) with the parameter -ax asm10 (ref. 97). SyRI (v.1.6) was used to detect the SVs with the default parameters⁴⁰. For PAVs, only deletions or insertions larger than 50 bp were retained. The intersect function in BEDTools (v.2.30.0) was used to calculate the TE ratios in the SV-involved segments⁹⁸.

Identification of A/B compartment switches

A/B compartments were identified using FAN-C with a 500-kb bin size. Inaccurately identified compartments were manually filtered out with the assistance of the Hi-C heat map and the AB eigenvector plot. Compartments smaller than 3 Mb were excluded from subsequent analysis. To identify A/B switches between *C. pubescens* and other assembled genomes, syntenic gene pairs in the A/B compartments were used as proxies, since direct bin comparisons were impractical due to size differences and extensive rearrangements. A bin in *C. pubescens* was classified as switched if over half of its genes had a different compartment status than their syntenic counterparts in another genome.

ATAC-seq, peak calling and ACR characterization

Young fresh leaves from 12 genome-assembled accessions and young fruits from *C. pubescens*, *C. baccatum*, *C. chinense*, *C. annuum* var. *glabriusculum* PM1533 and *C. annuum* 'Zunla-1' and 'Qiemen' were flash-frozen in liquid nitrogen. Two to three biological replicates were prepared per accession. Nuclei isolation followed a standard ATAC-seq protocol⁹⁹. The Nextera DNA Library Preparation Kit (Illumina) was used for the transposition. The libraries were purified and sequenced on the Illumina Novaseq 6000 platform. The 150-bp paired-end sequencing data were aligned to genomes using BWA (v.0.7.17-r1188) with the default parameters¹⁰⁰. The alignment files were sorted using SAMtools, and PCR duplicates were removed using the MarkDuplicates function in Picard (v.2.27.0)¹⁰¹. Alignments were filtered to retain properly paired fragments with MAPQ greater than 20. Genrich (v.0.6.1; <https://github.com/jsh58/Genrich>) was used for peak calling with parameters -j; -p, 0.01; -a, 100; -v; -f. The P values ranked at the top 5% were selected as the final cut-off, and Genrich was rerun with this P -value cut-off to predict the putative ACRs. Distances between ACRs and adjacent genes were called using Homer (v.5; <http://homer.ucsd.edu/homer/ngs/peaks.html>). To assess TE overrepresentation in ACRs, simulated ACRs matching length and density distributions were generated using the shuffle function in BEDTools. Repeat contents of real and simulated ACRs were then compared using the intersect function in BEDTools.

Comparison of ACRs across accessions

To assess ACR sequence conservation, the sequences of ACRs from each accession were aligned to the *C. pubescens* genome using BLASTn with $E < 1 \times 10^{-5}$, and hits with the top five scores were retained. ACRs overlapping genes were excluded. Using syntenic segment information, syntenic target regions in the *C. pubescens* genome were identified for query ACRs. If the top-scoring hit matched the syntenic region, it was retained as a valid 'syntenic hit' ('syn hit'). If not, subsequent hits were examined iteratively. If none of these hits matched the syntenic region, the top-scoring hit with identity and coverage higher than 80% was considered a valid 'jumping hit'; otherwise, the ACR was considered 'no hit'. Sequence conservation of ACRs was compared to those of randomly selected intergenic sequences with the same length distribution

using the shuffle function in BEDTools. The overlap of valid hits with *C. pubescens* ACRs was identified using the intersect function in BEDTools, and overlapping ACRs were defined as acACRs between the two genomes. For gene–ACR distance comparisons between *C. pubescens* and *C. rhomboideum*, only syntenic gene and ACR pairs were analysed, excluding gene-overlapping ACRs. Gene–ACR pairs with twofold distance changes (in the 5–50 kb range) were selected for further analysis. To compare repeat ratios in ACRs with different conservation levels, we generated several sets of *C. pubescens* ACRs according to their occurrence frequency across *Capsicum* accessions. Gene-overlapping ACRs were excluded to reduce the influence of repeat-sparse genic regions. The order of conservation from high to low as well as the methods for ACR set generation are detailed in Supplementary Table 15. The average repeat ratio in each ACR set was calculated.

Clustering of acACRs

The acACR pairs between genome pairs were clustered into acACR groups. Orthologous acACR pairs sharing at least one common member were grouped together. To address potential false-positive acACR pairs inflating group sizes, the Markov Cluster algorithm (MCL; inflation, 2.0) was used to refine groups¹⁰². MCL distinguishes densely connected orthologous groups from sparsely connected ones and splits sparsely connected groups into separate groups. Unclustered ACRs were considered specific ACRs, while acACR groups containing members from all investigated genomes were considered core acACRs. The acACR groups containing members from at least three *C. annuum* accessions (*C. annuum* var. *glabriusculum* included) but none of the non-*C. annuum* accessions were defined as *annuum*-specific acACRs.

Determination of the repeat/DNA methylation ratios

ACRs were partitioned into ten bins, while their upstream and downstream flanking sequences (5 kb long) were divided into 100-bp bins with a step size of 10 bp. The repeat ratio in each bin was calculated, and the ratios were then averaged among the ACR sets. A similar analysis was conducted to determine the ratio of *Gypsy* LTR-RTs surrounding genes. To explore DNA methylation levels around ACRs and genes, we used previously generated DNA methylation data from the leaf organ of ‘Zunla-1’ (PRJCA025503, Genome Sequence Archive, <https://ngdc.cncb.ac.cn/gsa/>), which was aligned to the ‘Zunla-1’ genome using Bismark (v.0.20.0)¹⁰³. Similarly, the 5-kb flanking regions of ACRs/genes were divided into 100-bp bins, while the body regions of ACRs/genes were partitioned into ten bins. The weighted methylation level was calculated for each bin.

Analyses of the influences of SVs on ACRs

C. pubescens ACRs were aligned to those of other *Capsicum* species, and hits were categorized as ‘syn’ or ‘jumping’ on the basis of prior criteria. Variants between *C. pubescens* and other accessions were identified using SyRI, with *C. pubescens* as the reference. *C. pubescens* ACRs showing sequence deletions (>50 bp, coverage >20%) in query genomes were considered to have PAVs; those with deletions <50 bp were considered to be without PAVs. The proportion of hit sequences occupied by PAVs was calculated using the coverage function in BEDTools. To evaluate the influence of location changes on the conservation of ACRs, hits without PAVs were retained to exclude the potential effects exerted by PAVs. The retained ‘syn’ and ‘jumping’ hits were considered the location-conserved and location-changed hits, respectively. To assess the influence of PAVs on ACR conservation, only ‘syn’ hits were analysed. For each type of hit, hits overlapping with ACRs in the corresponding accession were counted. To further estimate the positional influence of PAVs inside the ACRs on conservation, ACRs were partitioned into 20 bins centred on their predicted summits. Sequence deletion ratios for each bin were calculated and averaged for ACRs that retained or lost accessibility in the target species or accession.

Population resequencing and variant calling

A total of 124 accessions were selected according to their phylogenetic relationships in our previous resequencing study¹⁸. For simplification, the previously defined nine *C. annuum* groups were merged into six groups (Supplementary Table 12). DNA was isolated from the leaves of these accessions using the aforementioned method, and the constructed libraries were sequenced on a DNBSEQ-T7 platform. In total, approximately 16.27 Tb of paired-end data were generated. The deep resequencing data for these selected accessions as well as the DNA sequencing data used for polishing the assemblies were aligned to the PM1533 genome using BWA, and variants were called using the HaplotypeCaller function of GATK (v.4.1.4.0)¹⁰¹. The raw VCF file was filtered using VCFtools (v.0.1.6) with the parameters `-mac, 3; -minQ, 30; -max-missing, 0.9; -remove-indels; -min-meanDP, 10; -max-meanDP, 50; -minDP, 5; -maxDP, 80; -maf, 0.03` (ref. 104).

Construction and analysis of the haploblock map

To guide accession selection for accurately detecting introgressions on the basis of variants, potential interspecies introgressions were initially estimated using genome assemblies. Specifically, genes from PM1533 were BLASTn searched against non-*C. annuum* species, excluding *C. rhomboideum*. For each gene in PM1533, the best hit with an identity higher than 99.8% was identified, and the number of best hits were counted for each species within each 1-Mb bin. Bins containing more than five genes were considered potential introgressed bins if over 60% of the best hits originated from the same species. Since the majority of potential introgressed bins (255/276, 92%) were derived from *C. frutescens* or *C. chinense*, we selected 109 accessions from *C. frutescens*, *C. chinense* and *C. annuum* for downstream analyses. We used Introblocker (v.1.0.1) to identify the haploblocks with a bin size of 1 Mb in each accession using the semi-supervised mode. Introblocker was run with the parameters `cluster_thr = 3,000` and `smooth_thr = 2,000` to identify interspecies haplotypic differentiation and `cluster_thr = 1,000` and `smooth_thr = 600` to identify intraspecies haplotypic differentiation⁵¹. Ancient introgressions from *C. frutescens* or *C. chinense* to *C. annuum* were defined as haploblocks present in >80% of *C. annuum* accessions. Ancestral undifferentiated haploblocks—frequent in both *C. frutescens* and *C. chinense* (>80%)—were excluded. Meanwhile, haploblocks found exclusively in *C. frutescens* or *C. chinense* (with no limits on accession number) were defined as *C. frutescens*- or *C. chinense*-contributed haploblocks. Genetic distances between accessions were calculated using plink embedded in Introblocker to examine whether *C. frutescens*-contributed haploblocks were derived from recent gene flow from *C. annuum* to *C. frutescens*. Distances between *C. annuum* and *C. frutescens* were compared to intra-*C. annuum* distances using Student’s *t*-test ($P < 0.05$). Chromosomal haploblock diversity and transitions were evaluated by partitioning chromosomes into 20 bins centred on centromeres. Haploblocks dominantly found in blocky fruit peppers were screened on the basis of high frequency in blocky groups (≥ 12 accessions, 75%) and low frequency in other groups (≤ 16 accessions, 25%).

RNA-seq and data processing

For the RNA-seq assay paralleling the ATAC-seq, RNA was isolated from the same samples used for ATAC-seq. Libraries were constructed using the VAHTS mRNA-seq v.2 Library Prep Kit (Illumina) and were sequenced on the Illumina Novaseq 6000 platform. To investigate the transcription of the *Up* gene in *C. chinense* accessions with different fruit orientations, the pedicel and placenta tissues were collected and flash-frozen in liquid nitrogen. Libraries were constructed using the Hieff NGS Ultima Dual-mode mRNA Library Prep Kit for Illumina and were sequenced on the DNBSEQ-T7 platform. The 150-bp paired-end data were aligned to the corresponding genomes using HISAT2, and gene quantification was conducted using featureCounts (v.2.0.3)^{81,105}. TPMs calculated from each replicate were averaged as the gene expression levels.

Phylogenetic analysis and validation of SVs

To distinguish the genotypes of the *L* gene in *C. annuum* and *C. chinense* accessions, the resequenced data were aligned to the *C. annuum* ‘Qiemen’ genome, since the genomic segment carrying the *L* gene was absent in PM1533 due to the PAV. After the accessions without the segment were excluded, variants in VCF format were called using the mpileup function in SAMtools and the call function in BCFtools (v.1.13). The VCF file was converted to the PHYLIP alignment format, and a phylogenetic tree was constructed using IQ-TREE (v.2.1.4)¹⁰⁶. The same methods were used to construct the phylogenetic tree of the haploblock carrying *Pun1* (152–153 Mb in Chr02). To examine resistance to the P_{1,2} pathotype of PMMoV, plants were mechanically inoculated by rubbing the virus onto two true leaves with Carborundum –400 mesh when three or four leaves were fully expanded. After inoculation, the pepper seedlings were kept in a greenhouse maintained at 25–30 °C. To validate the SVs associated with the *Up* and *CCS* genes, primers spanning the variations were designed (Supplementary Table 16), and the genotypes were determined by the PCR-amplified fragment sizes.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All sequencing data generated in this study have been deposited at the Genome Sequence Archive (<https://ngdc.cncb.ac.cn/gsa/>) under BioProject PRJCA019635. The genome assemblies with annotations are available at the Genome Warehouse (<https://ngdc.cncb.ac.cn/gwh/>) under the same BioProject ID, and the variation data are available via the Genome Variation Map (<https://ngdc.cncb.ac.cn/gvm/>) with accession number GVM000925. These assemblies can also be accessed at SolPGD (http://www.bioinformatics-lab.cn/files/genomes/pepper_pan).

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Author contributions

F.C. and L.W. conceived and designed the research. H.Y., Y.C., Z.Z. and Y.J. collected the samples. K.Z. and L.Z. analysed the data. H.Y. conducted the validation experiments. X.L., Y.M., X.W., T.L., W.F., Y.W. and S.C. participated in the data analyses and visualization. C.G., D.Z. and H.W. investigated the agricultural characteristics. K.Z., F.C., H.Y. and L.Z. wrote the manuscript. F.C., K.Z. and L.W. revised the manuscript. All the authors reviewed and approved the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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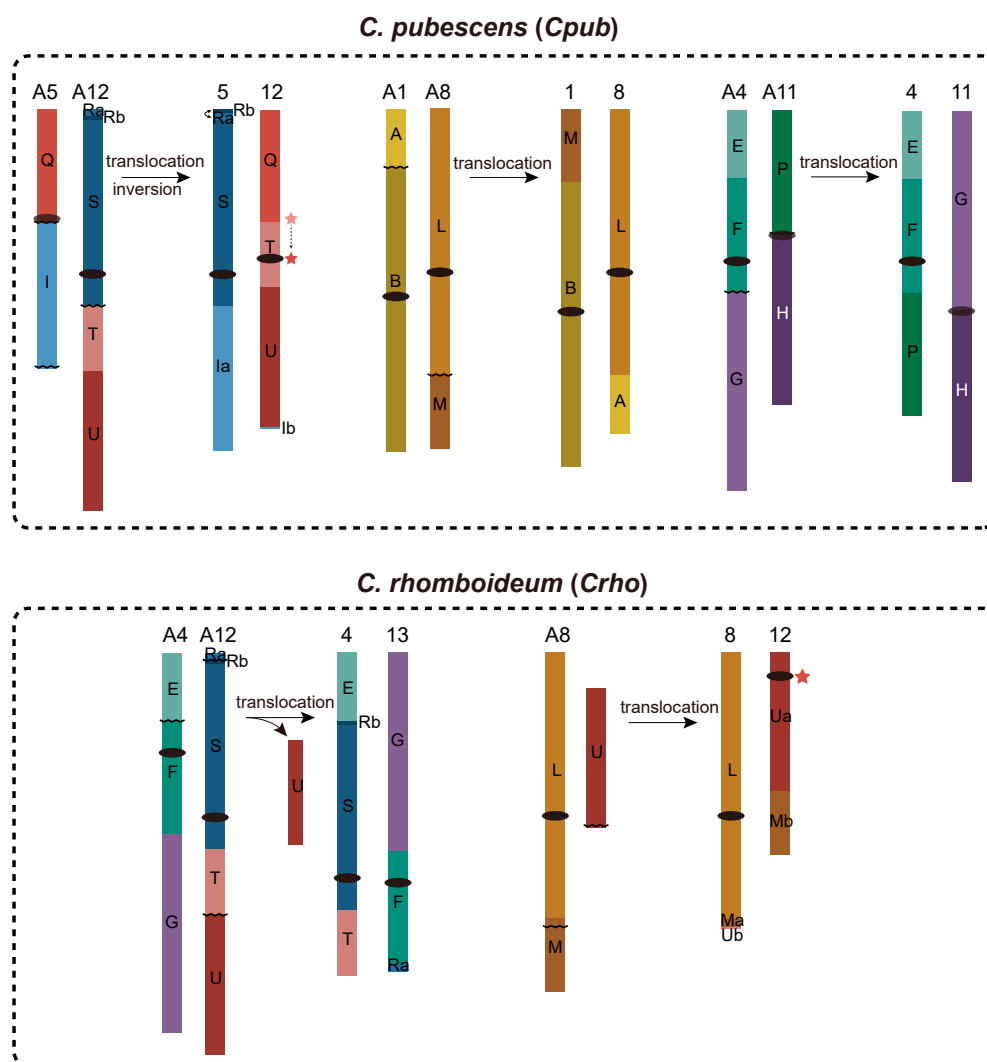
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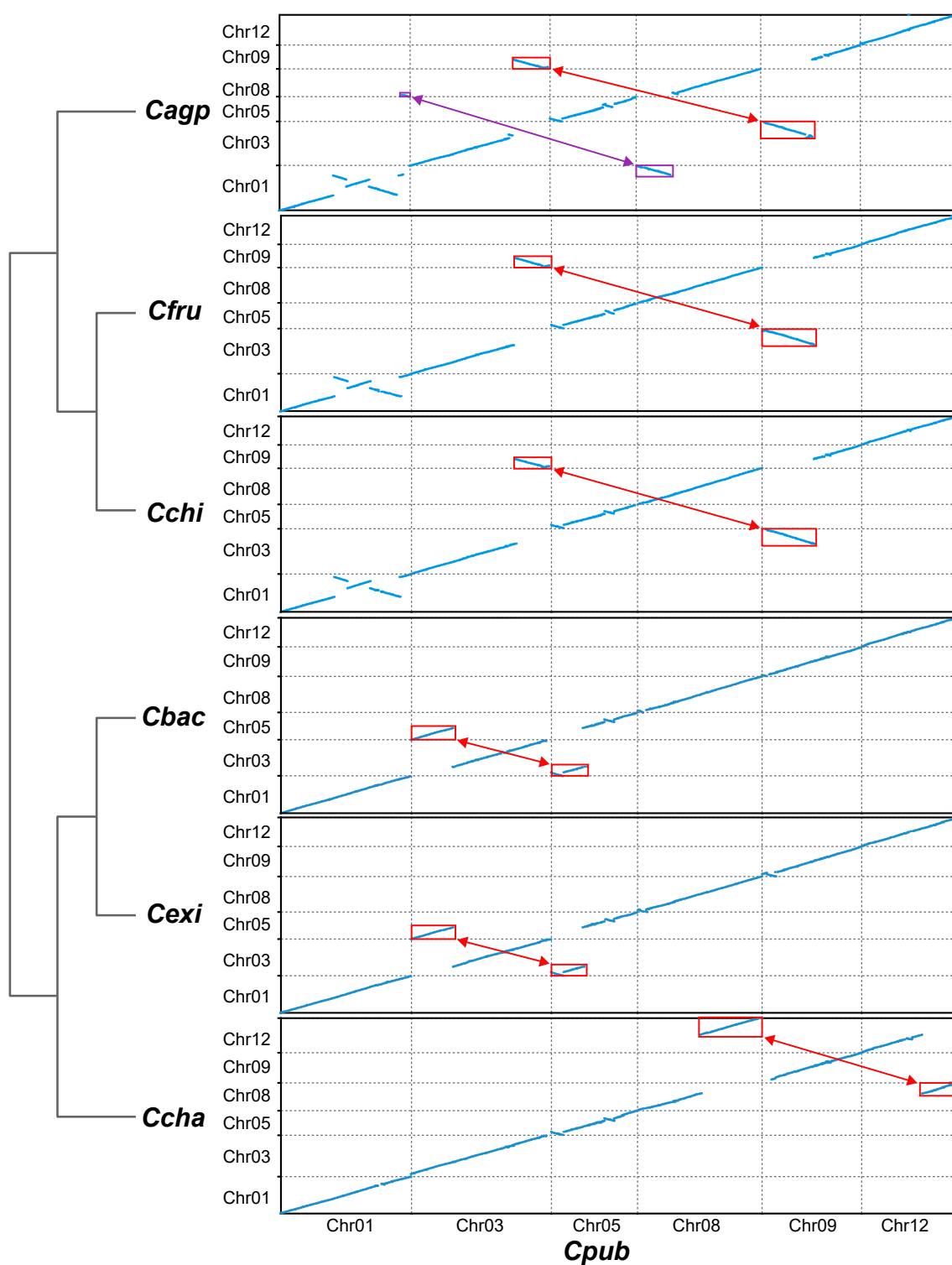
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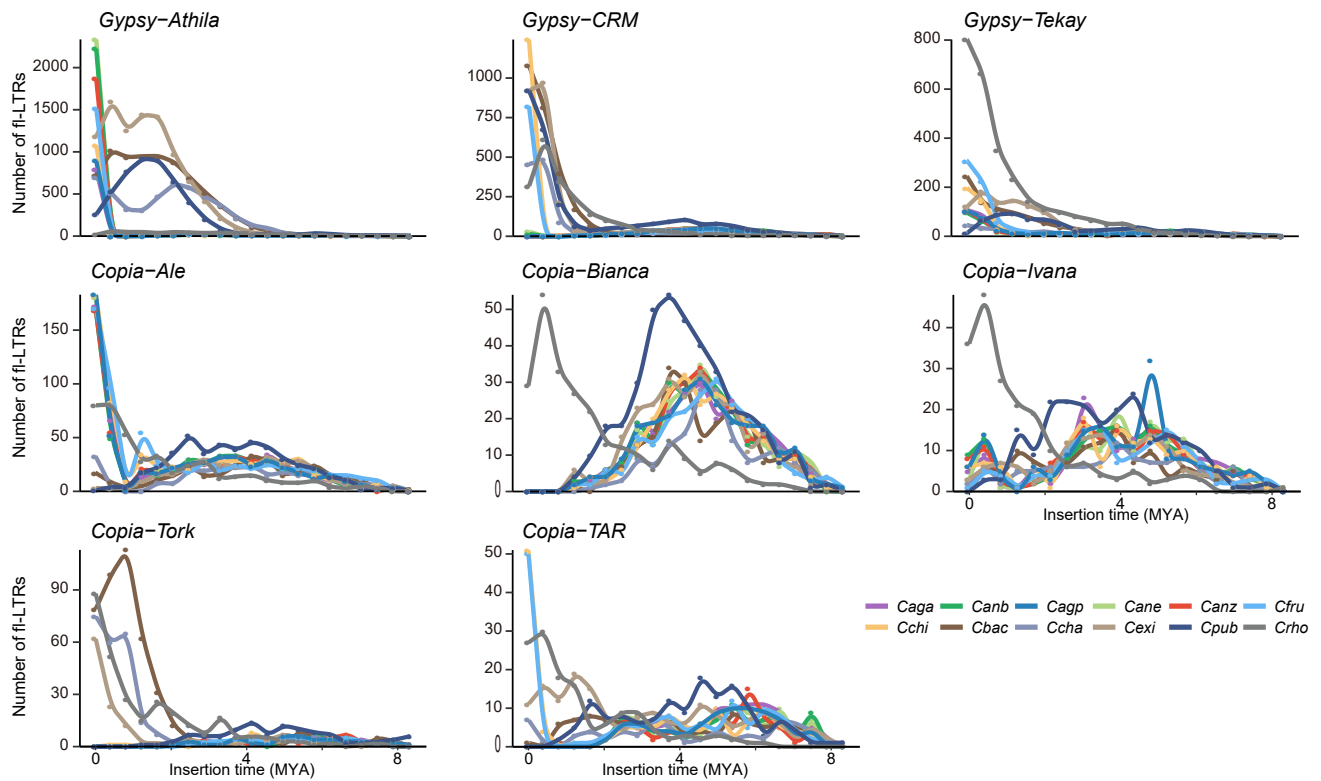


Extended Data Fig. 1 | Karyotype evolution of *C. pubescens* and *C. rhomboideum* following divergence from their common ancestor. Colored boxes represent genomic blocks. Red stars indicate changes in centromere positions or the emergence of new centromere. Broken lines denote potential breakpoints.

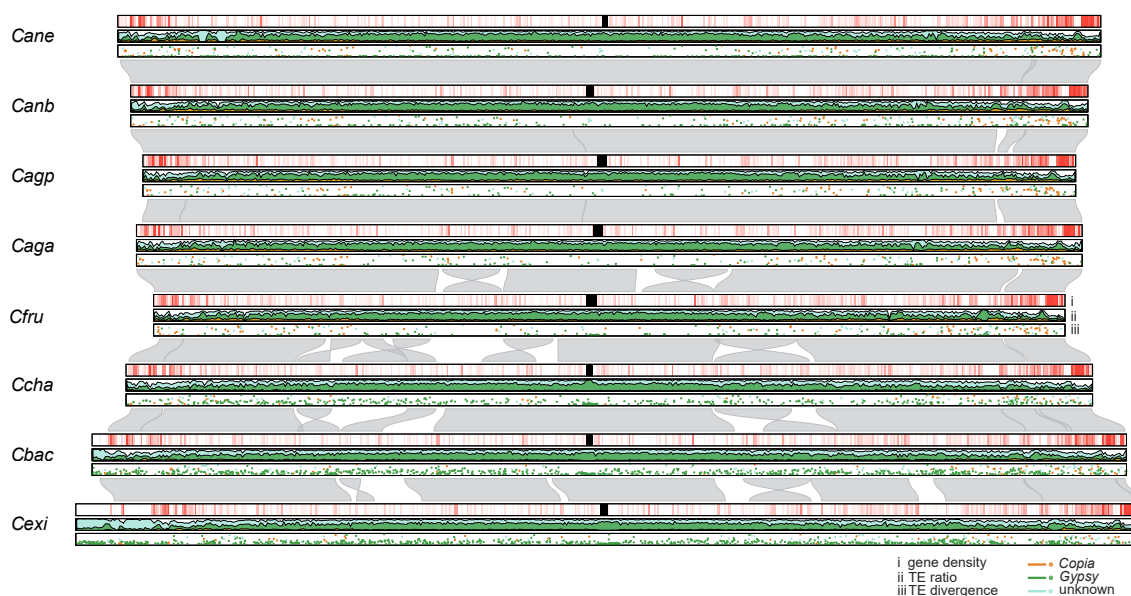


Extended Data Fig. 2 | Syntenic plots showing interchromosomal translocations in *Capsicum* genomes compared to *C. pubescens*. Genomes from phylogenetic clusters one and two were compared to the genome of

C. pubescens. Chromosomes involved in translocations are displayed, with potential interchromosomal translocations highlighted by red or purple boxes. Species and accession abbreviations are as defined in Fig. 1.

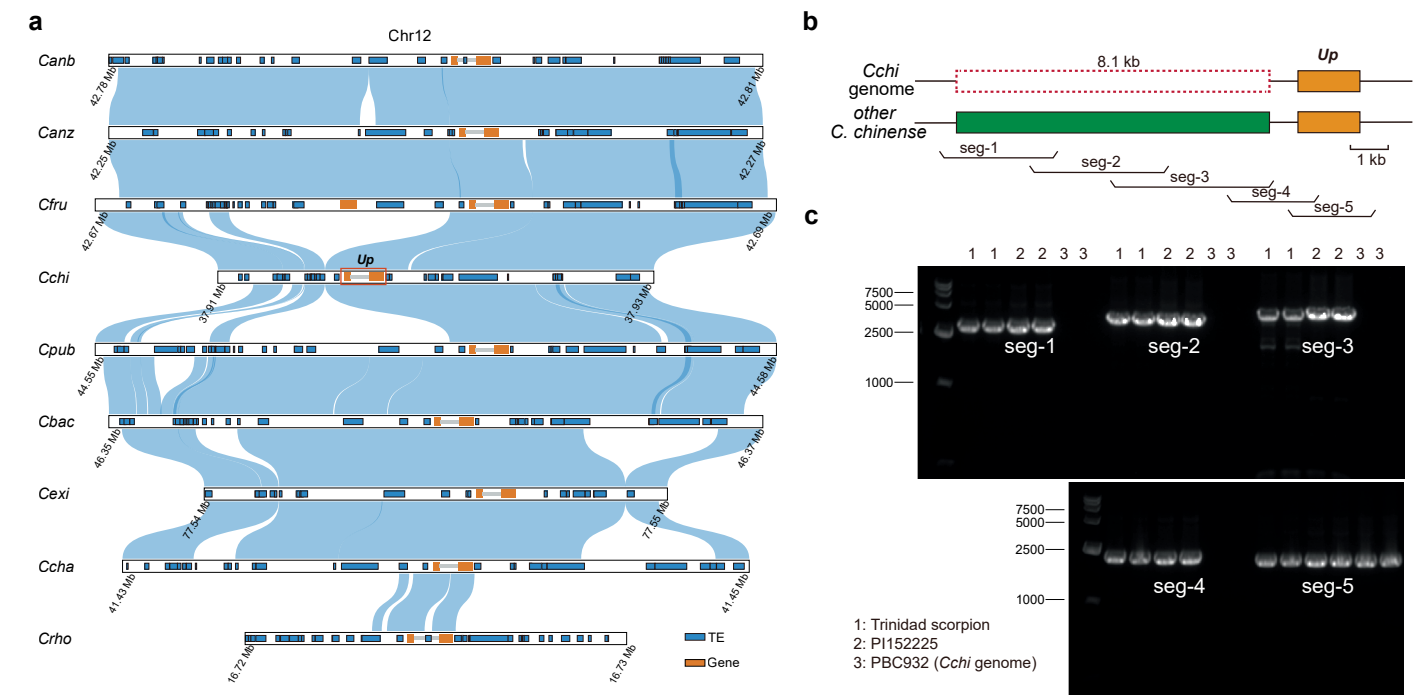


Extended Data Fig. 3 | Insertion time distribution of full-length retrotransposons (fl-LTRs) at the family level. Insertion times were estimated based on sequence divergence between the long terminal repeats (LTRs). The three most abundant *Gypsy* families and the five most abundant *Copia* families are shown. MYA: million years ago.



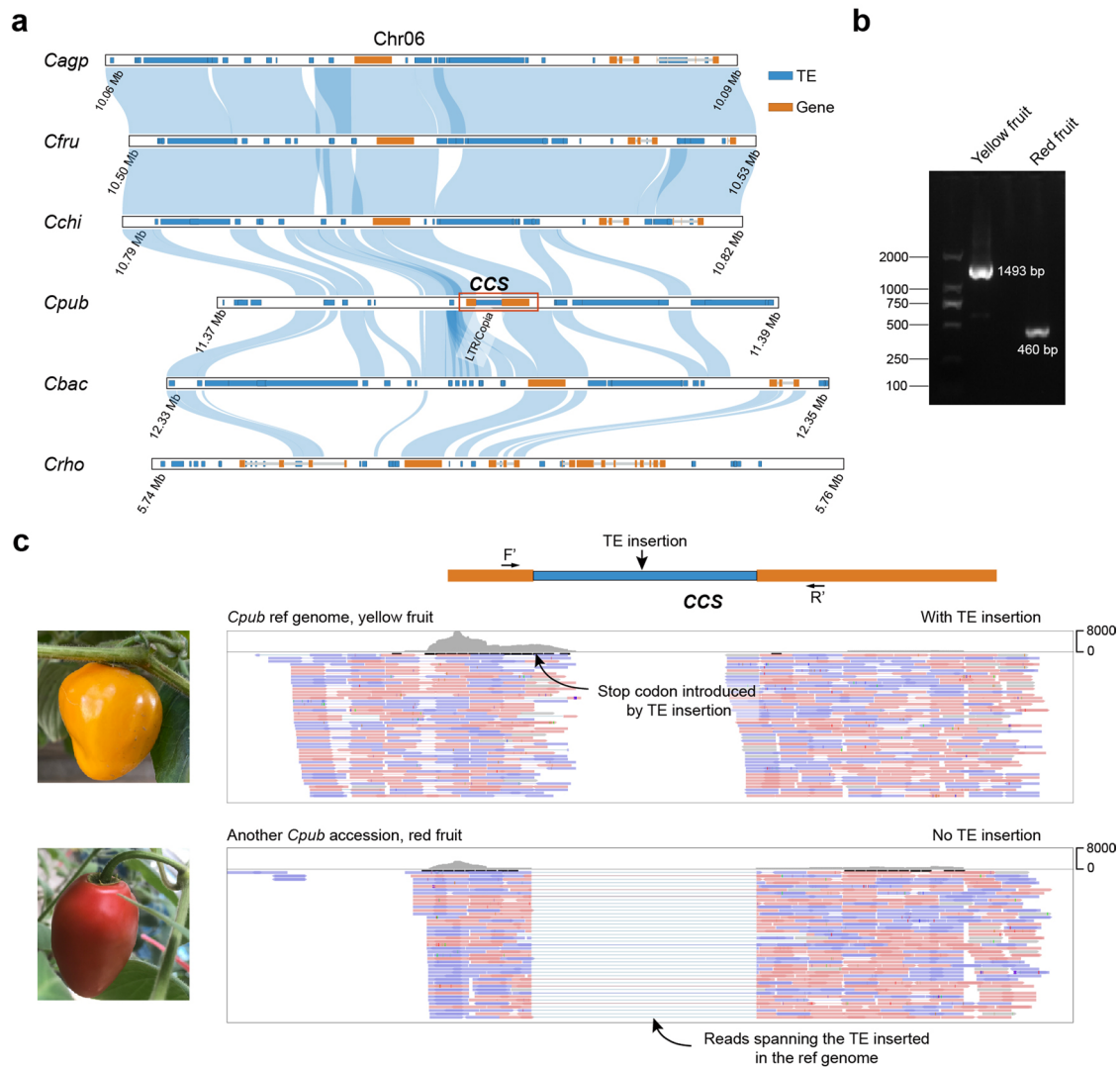
Extended Data Fig. 4 | Multiple genomic alignments of *Capsicum* genomes with the Chr07 as an example. Dark blocks denote the centromeres. In the TE ratio (ii) and TE divergence (iii) rows, the orange, green, and teal blue filled lines/dots denote the full-length *Copia*, *Gypsy*, and other unknown types of TEs,

respectively. The range of the TE divergence (iii) row is from 0 to 0.1, denoting the divergence between two LTRs. Eight genomes are shown here, while the remaining four genomes are shown in Fig. 3c.



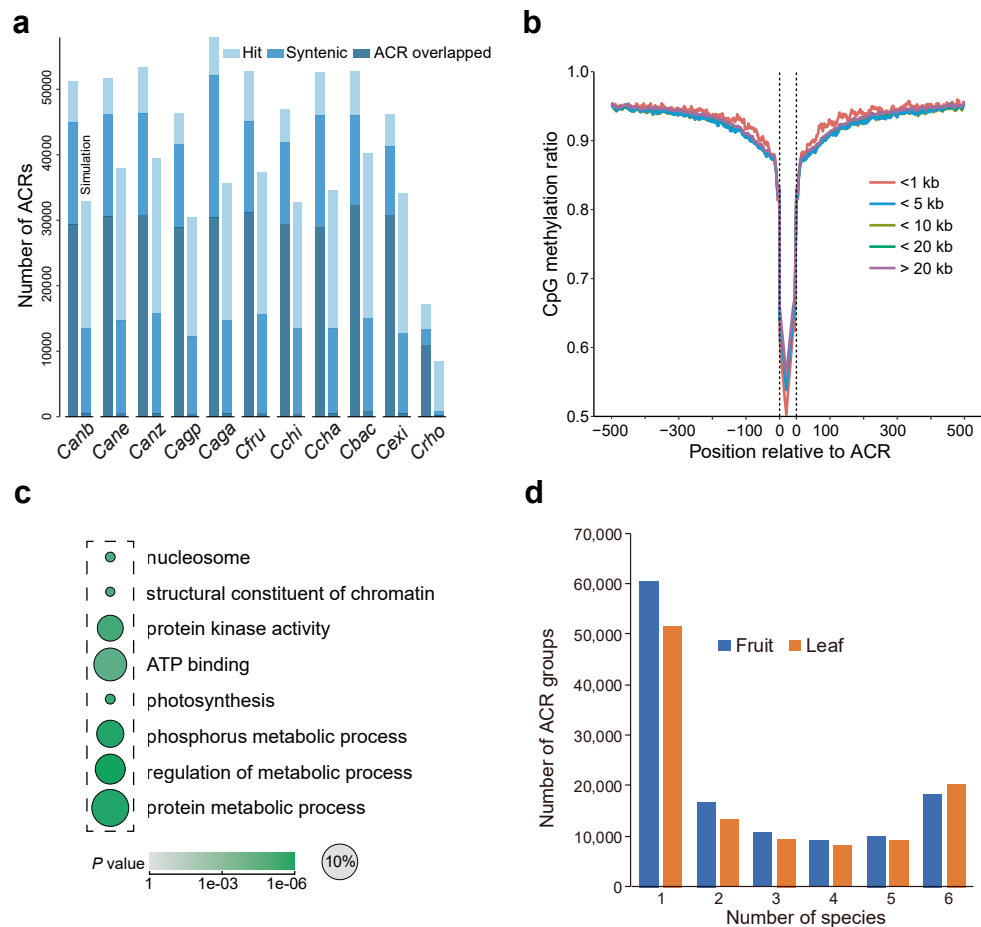
Extended Data Fig. 5 | Validation of the large structural variation in the *Up* gene of *C. chinense*. **a**, *Up*-related multiple genomic alignments among *Capsicum* genomes. **b**, Diagram showing the relative position of the structure variation and designed primers. The deletion in the corresponding genome is denoted

by a dashed-line box. **c**, PCR amplification results validating the large structural variation. Three *C. chinense* accessions were tested, with the third accession corresponding to the one with an assembled genome.



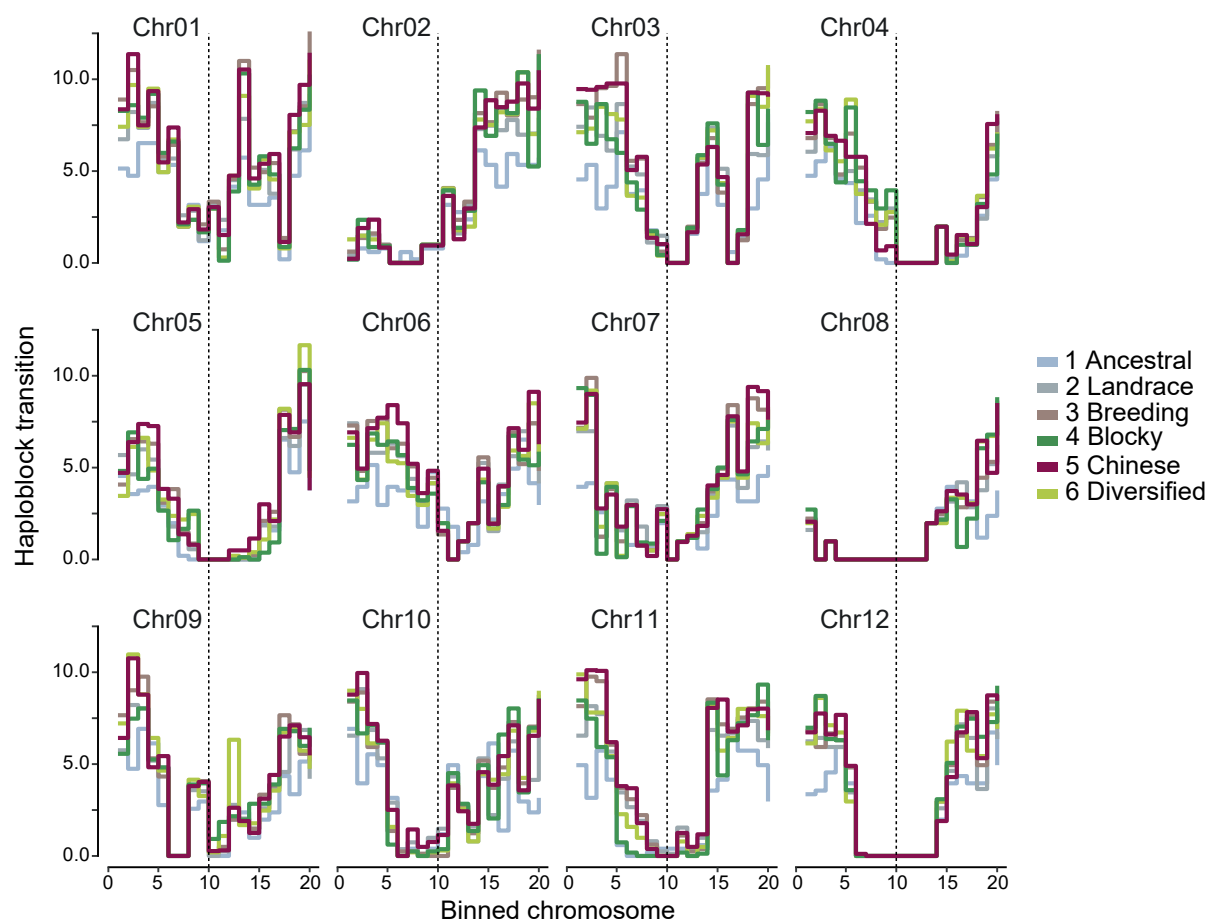
Extended Data Fig. 6 | A TE insertion in the gene body of *CCS* in the genome of *C. pubescens* is closely associated with yellow fruit color. **a, *CCS*-related multiple genomic alignments among *Capsicum* genomes. Yellow boxes connected by grey bars denote the gene structure, while blue boxes denote**

transposable elements (TEs). **b**, PCR amplification validated the TE insertion, with primer locations shown in **c**. **c**, RNA-seq read alignments reveal that the TE insertion introduced a premature stop codon.

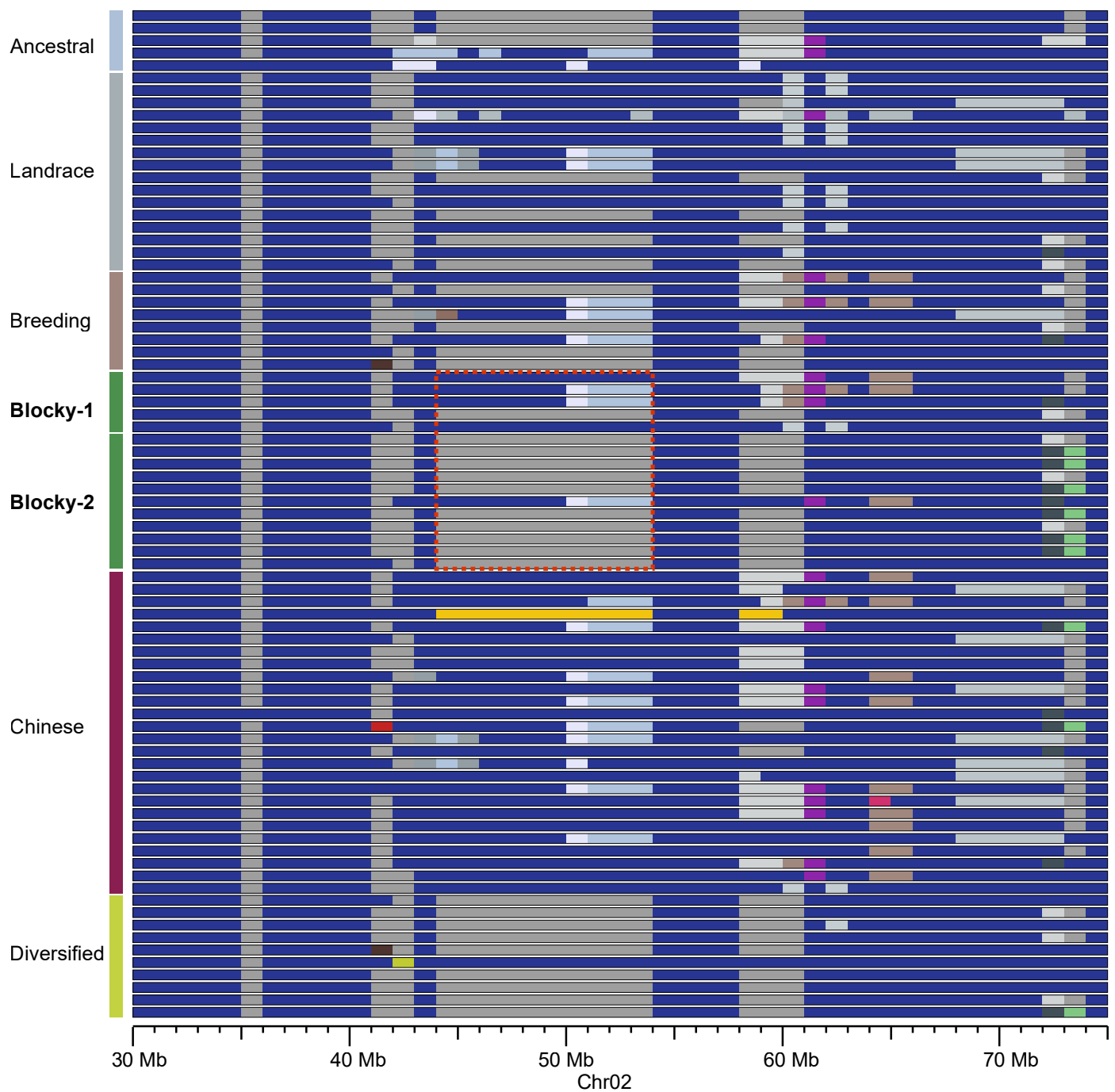


Extended Data Fig. 7 | Conservation of ACRs in *Capsicum* genomes.
a, Conservation levels of ACRs aligned to the *C. pubescens* genome. Hits located in syntenic regions are labeled as 'syntenic', while those overlapping with *C. pubescens* ACRs are labeled as 'ACR overlapped'. **b**, CpG methylation levels in and surrounding ACRs located at varying distances from their adjacent genes. **c**, GO enrichment analysis of genes with extremely conserved ACRs locating at

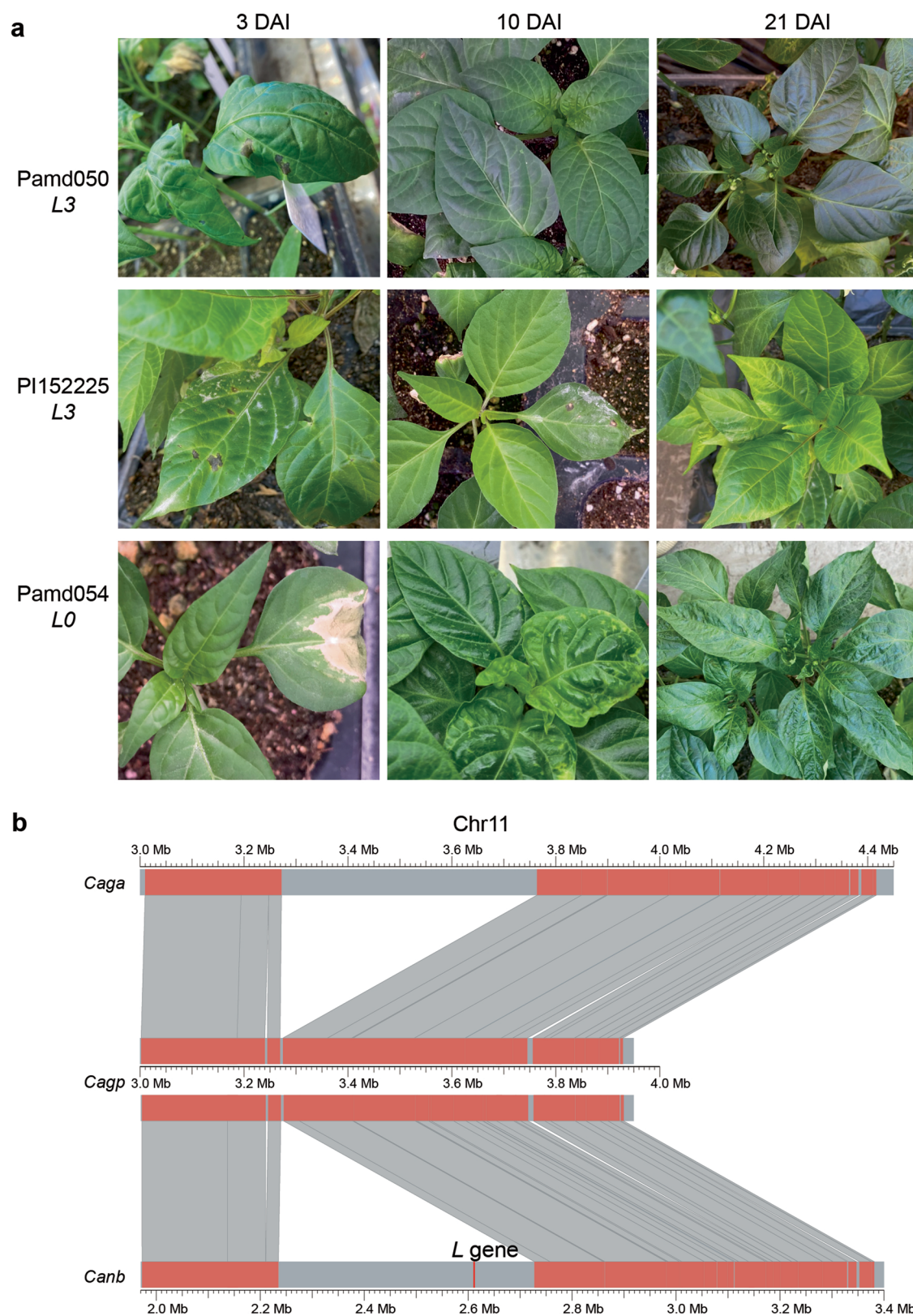
adjacent regions. **d**, Conservation level of fruit ACRs across *Capsicum* genomes, evaluated by the size of ACR groups. ACRs shared among all six genomes are considered highly conserved, while accession-specific ACRs are the least conserved. For comparison, leaf ACR groups were identified only using genomes with fruit ATAC-seq data rather than using all 12 *Capsicum* genomes.



Extended Data Fig. 8 | Haploblock transition frequency in each *C. annuum* group. Chromosomes were partitioned into 20 equal bins, with centromeres centered and indicated by dashed lines.



Extended Data Fig. 9 | Haploblock composition of the centromeric region in Chr02. The six *C. annuum* groups are indicated by colored bars on the left. The ancestral centro-haploblock (44–54 Mb), which was nearly fixed in blocky groups (blocky-1 and blocky-2), is highlighted by a dashed-line red box.



Extended Data Fig. 10 | PMMoV inoculation test and the *L* gene-related genomic alignments between different *C. annuum* genomes. **a, PMMoV inoculation test of the *L*³-carrying blocky accession (Pamd050) and *C. chinense* accession (PI152225), as well as the *L*⁰-carrying blocky accession (Pamd054) at three, 10, and 21 days after inoculation (DAI). **b**, *L* gene-related genomic**

alignments between different *C. annuum* genome. A large presence-absence variation (PAV) of 493 kb was found. Genomes with PAV presence: Ac1979 (*Caga*) and 'Qiemen' (*Canb*); Genomes with PAV absence: PM1533 (*Cagp*), 'Zunla-1' (*Canz*), and C5 (*Cane*).

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All sequencing data generated in this study have been deposited at the Genome Sequence Archive (GSA, <https://ngdc.cncb.ac.cn/gsa/>) under BioProject

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Sample size	No sample size calculation was performed. A total of 11 representative Capsicum accessions, including five wild and six domesticated accessions, were selected for genome assembly. This selection was guided by their phylogenetic relationships, ensuring they effectively represent the evolutionary diversity across Capsicum genus.
Data exclusions	No data were excluded from the analyses.
Replication	Two to three biological replicates were prepared per accession for ATAC-seq and RNA-seq assays. All replications were successful and were used.
Randomization	The sampling process for genomic DNA and RNA sequencing was conducted randomly.
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